

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 225 factors, ordered by in-sample L–S CAGR.

HOW TO READ

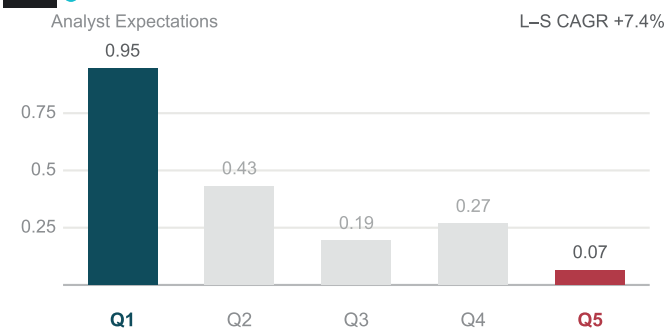
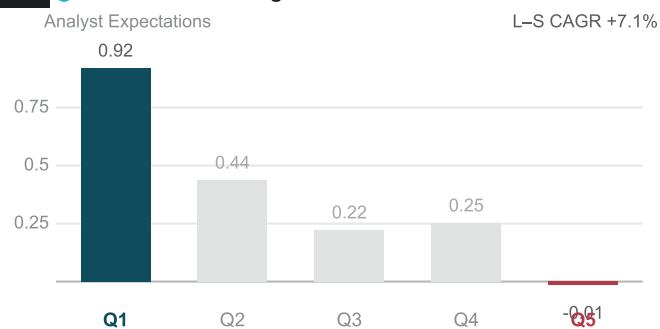
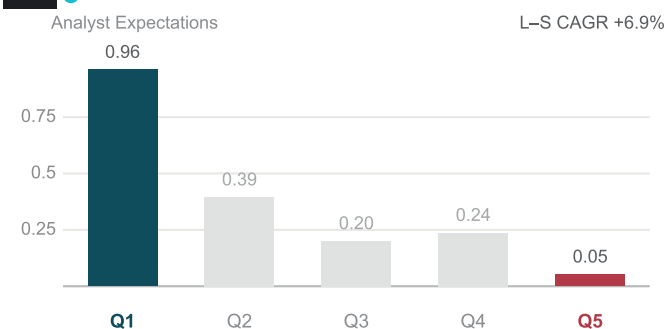
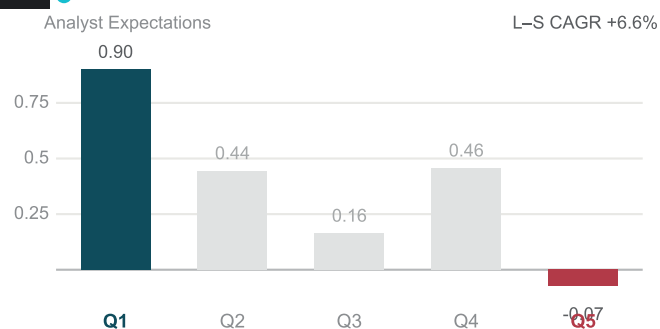
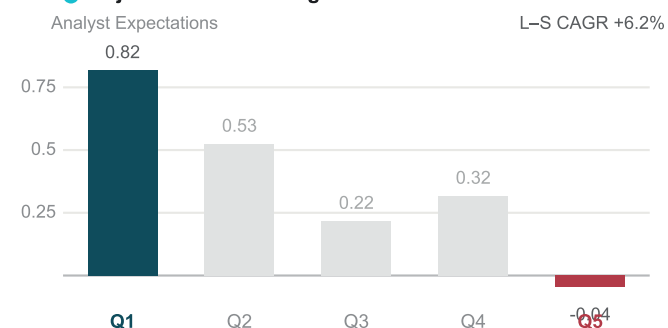
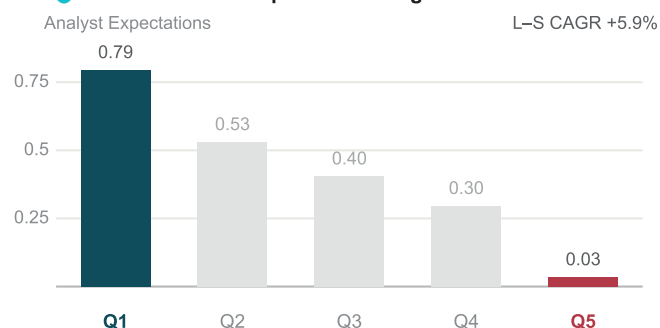
 Long  Short  Neutral

Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

FACTOR STYLES

 Valuation	35	 Price Momentum	50
 Earnings Quality	46	 Size	4
 Analyst Expectations	18	 Historical Growth	38
 Capital Efficiency	27	 Volatility	7

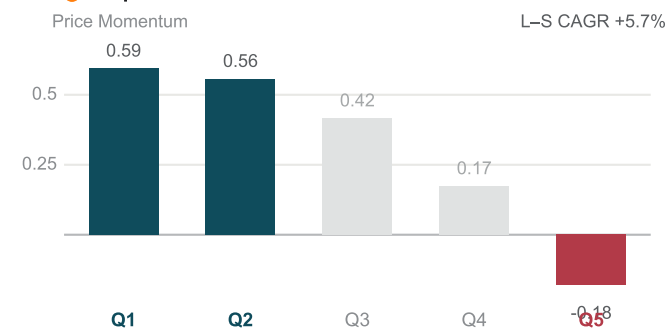
 001 In current model portfolio (28 factors)

001 3M Revision in FY2 EPS Estimate**002 Street Revision Magnitude****003 3M Revision in FY1 EPS Estimate****004 Number of EPS FY1 Revisions****005 Adjusted Revision Magnitude****006 Standardized Unexpected Earnings**

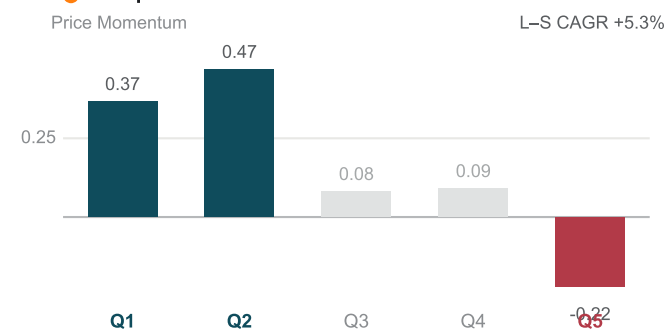
007 ● 6M Chg in Target Price



008 ● Slope of 52W Price Trend



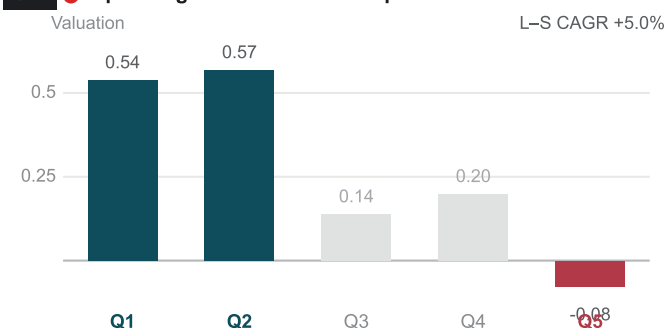
009 ● Sharpe Ratio



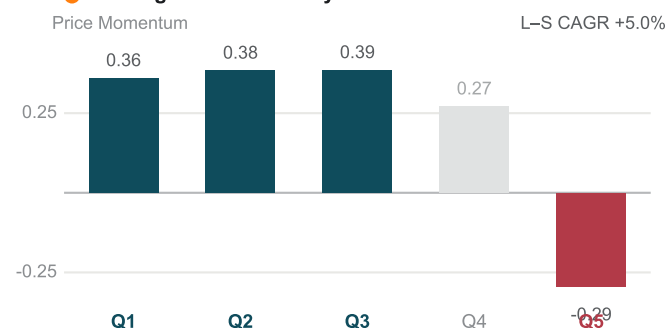
010 ● Cash Flow Return on Invested Capital

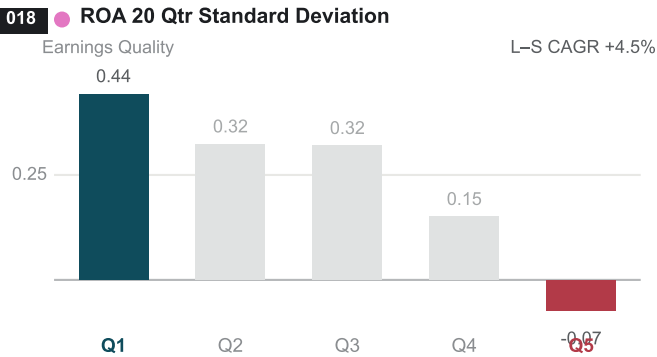
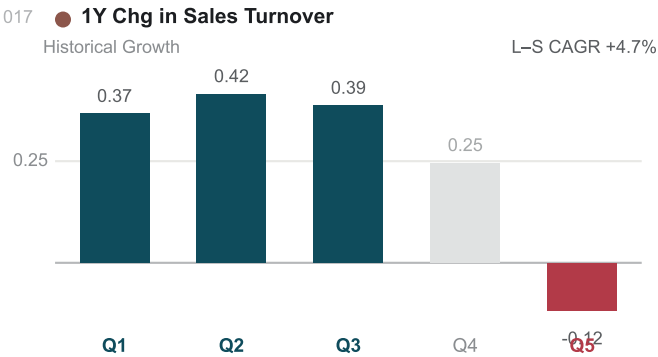
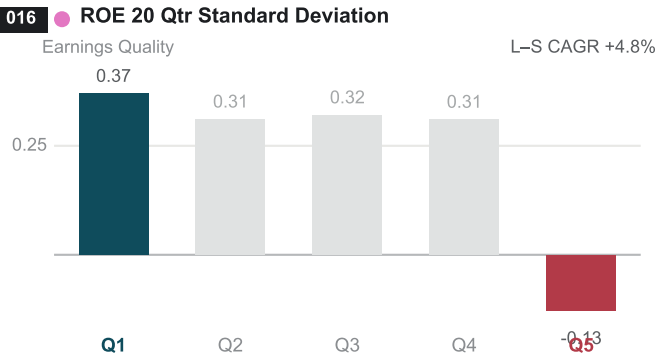
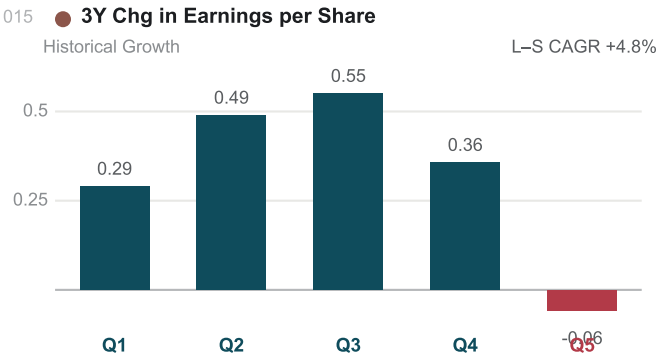
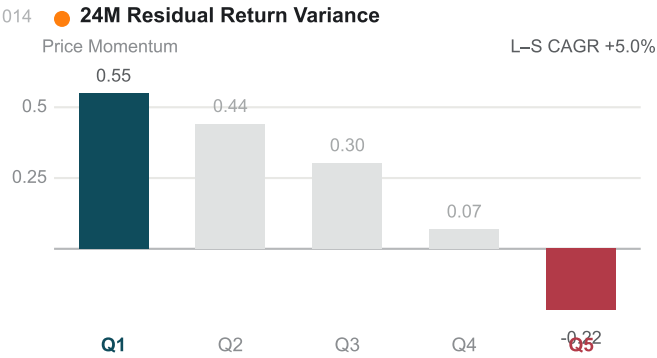
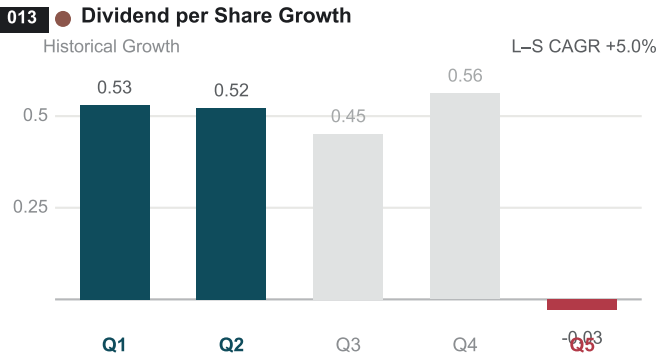


011 ● Operating Cash Flow to Enterprise Value



012 ● Closing Price to 260 Day Low

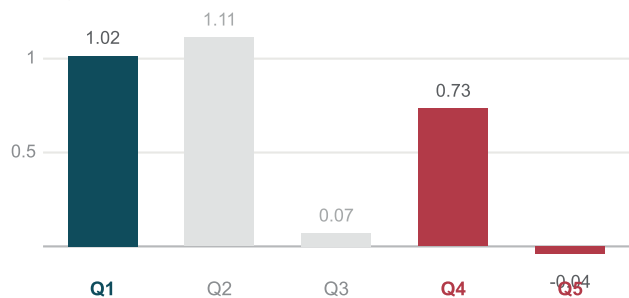




019 ● **Adjusted Number of EPS FY1 Revisions**

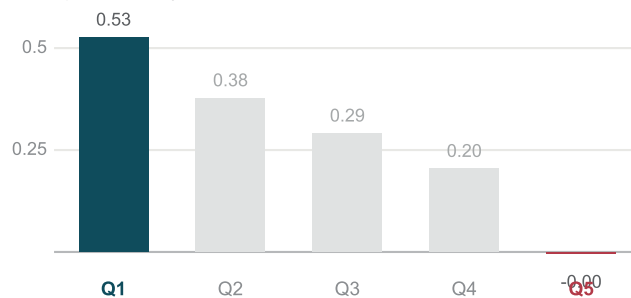
Analyst Expectations

L-S CAGR +4.5%

020 ● **Operating Cash Flow to Equity**

Capital Efficiency

L-S CAGR +4.4%

021 ● **Free Cash Flow to Sales**

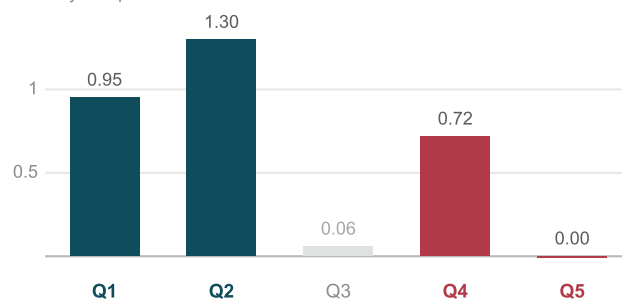
Earnings Quality

L-S CAGR +4.3%

022 ● **Analyst Earnings Estimate Diffusion**

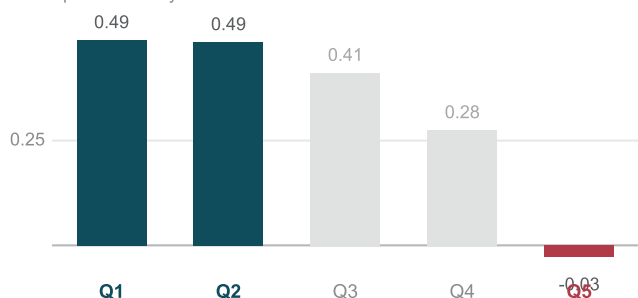
Analyst Expectations

L-S CAGR +4.3%

023 ● **Operating Cash Flow Ratio**

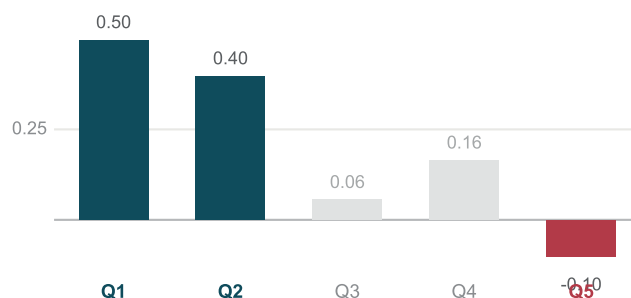
Capital Efficiency

L-S CAGR +4.3%

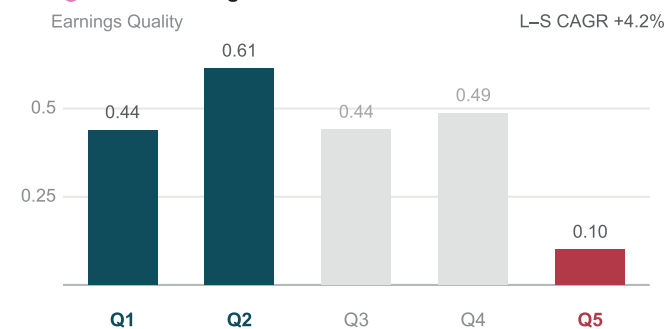
024 ● **Sales to Price Ratio**

Valuation

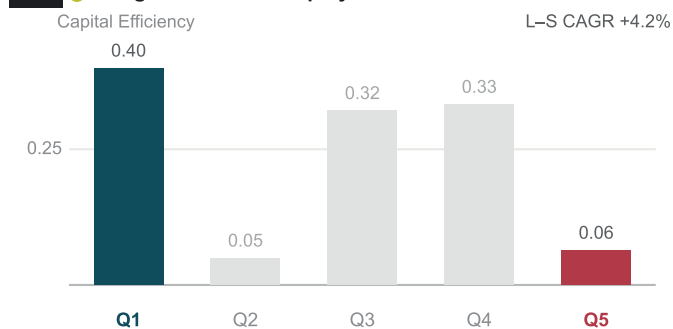
L-S CAGR +4.3%



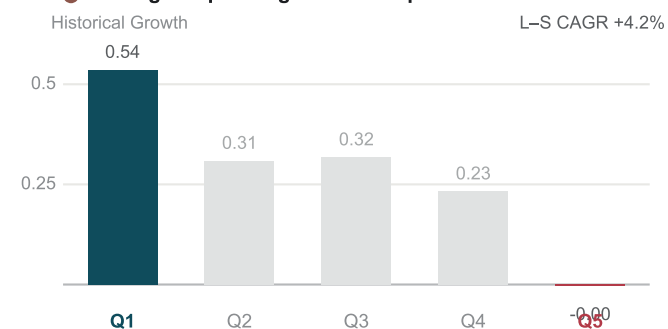
025 ● Net Profit Margin



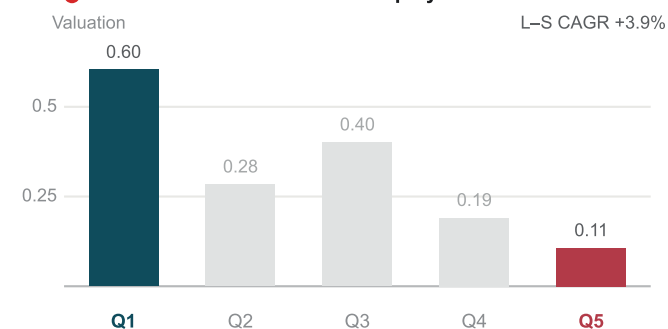
026 ● Long Term Debt to Equity Ratio



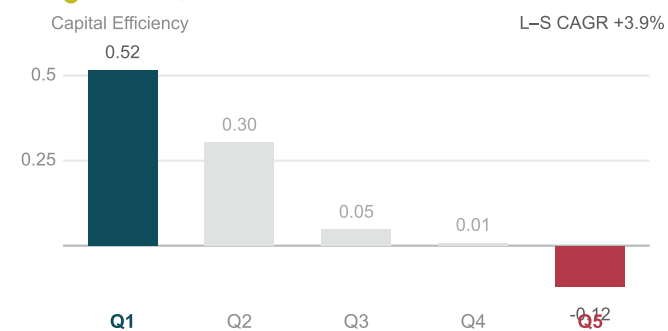
027 ● 1Y Chg in Operating Cash Flow per Share



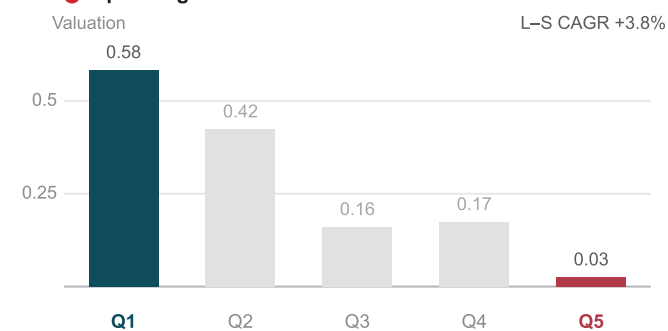
028 ● Free Cash Flow to Common Equity

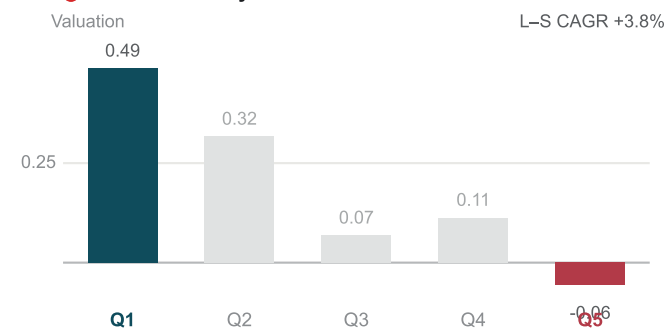
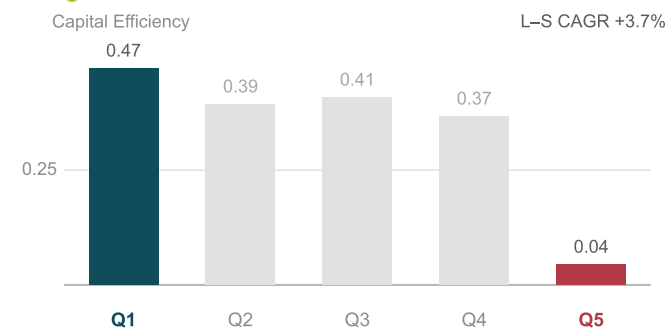
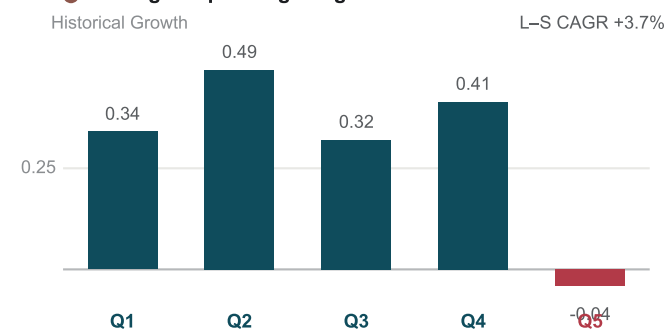
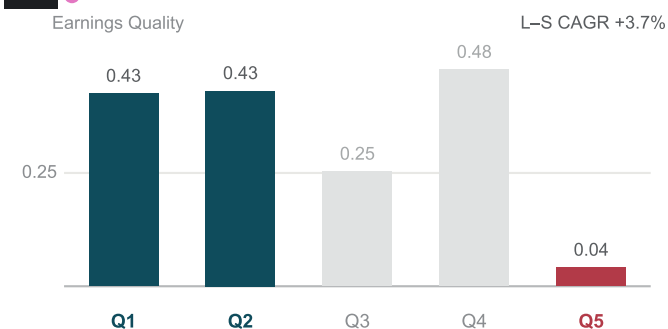
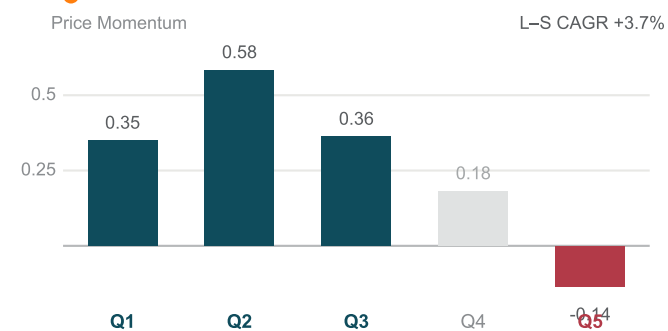
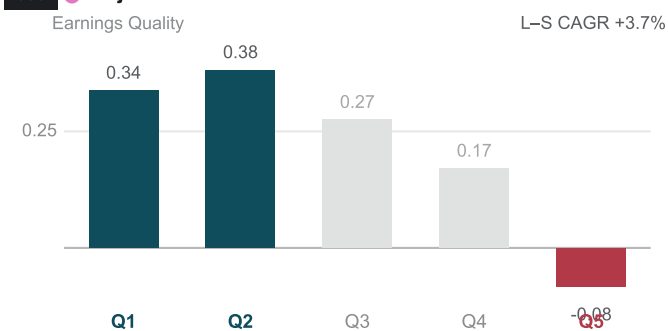


029 ● Tobin's Q Ratio

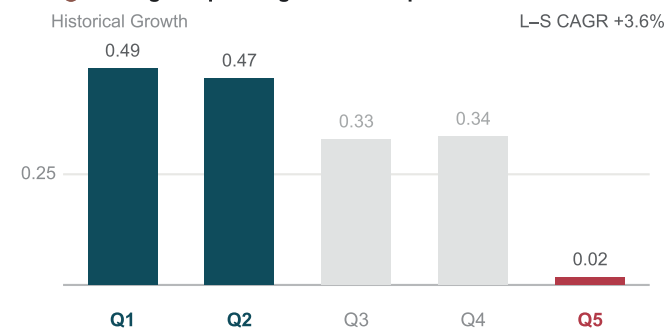


030 ● Operating Cash Flow to Price

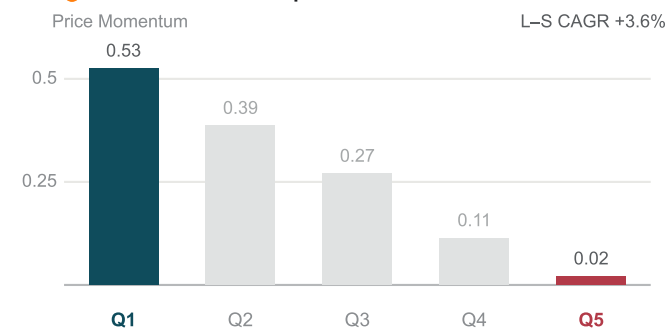


031 ● **Current Liability to Price**032 ● **Gross Profit to Assets**033 ● **3Y Chg in Operating Margin**034 ● **Asset Turnover**035 ● **180D Price TStat**036 ● **Adjusted Accruals**

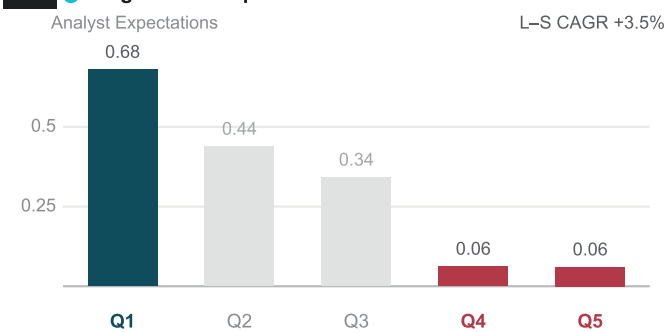
037 ● 3Y Chg in Operating Cash Flow per Share



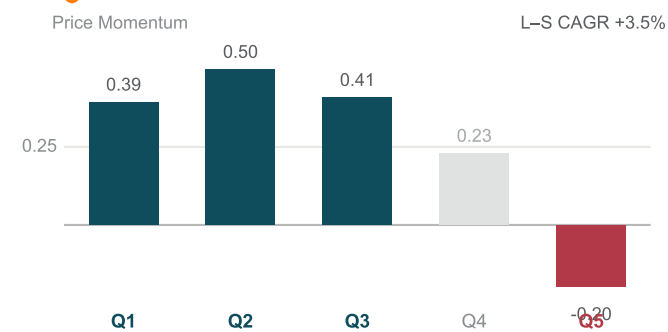
038 ● Institution Ownership Concentration



039 ● Target Price Gap to 6M EMA



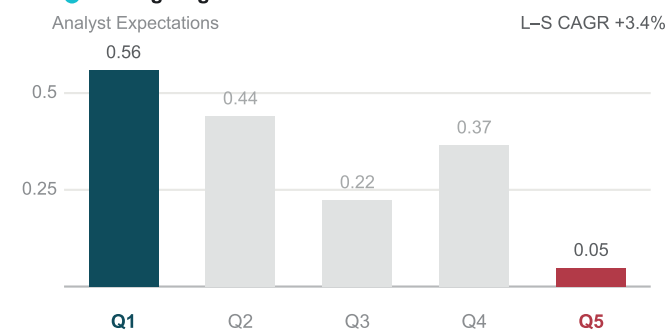
040 ● Share Turnover



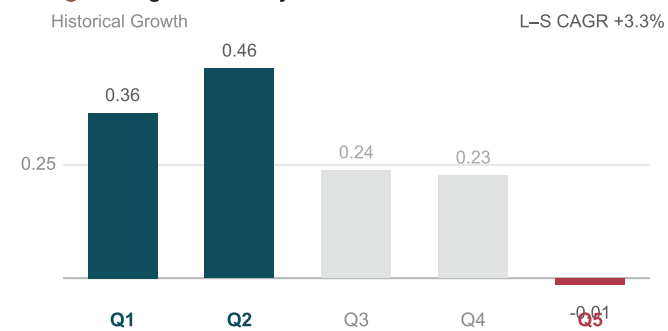
041 ● 1Y Chg in Sales to Price



042 ● 6M Avg Chg 1M Recommendation



043 ● 1Y Chg in Asset Adjusted EPS



044 ● 3Y Chg in Free Cash Flow per Share



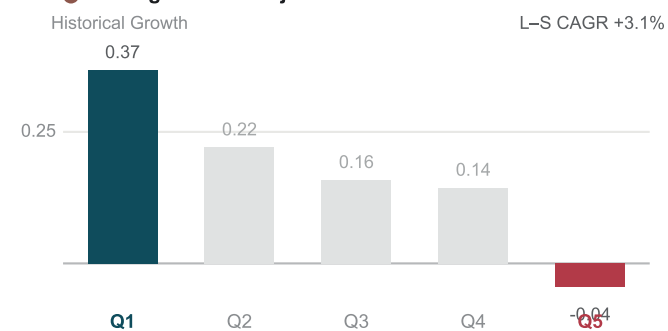
045 ● Capital Acquisition Ratio



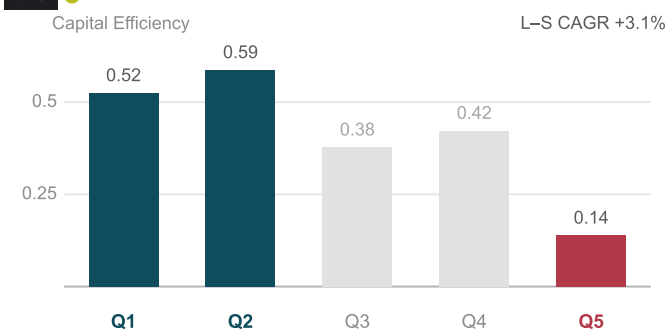
046 ● EBIT Margin



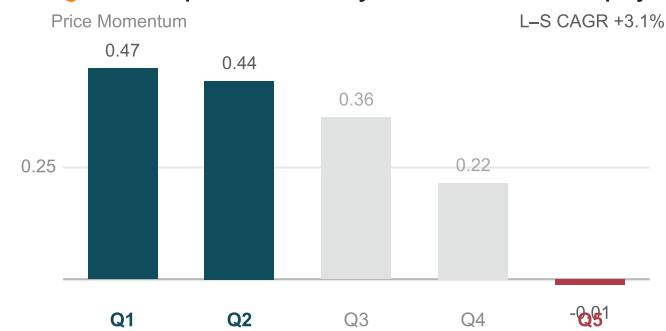
047 ● 1Y Chg in Asset Adjusted Free Cash Flow



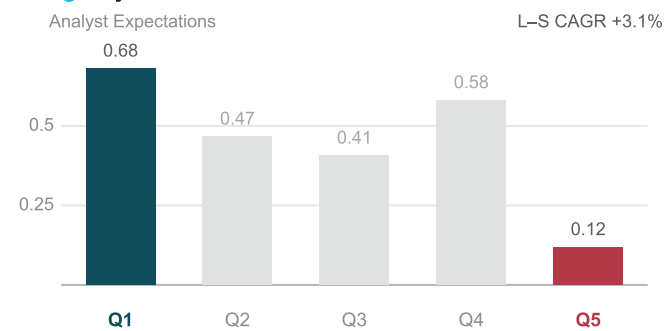
048 ● Receivables Turnover Ratio



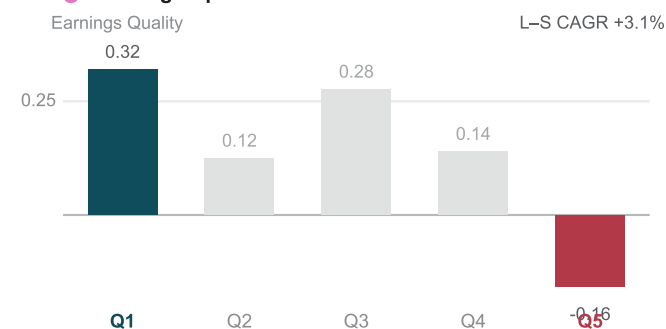
049 ● Ownership Breadth Stability of Institution Ownership by Hol...



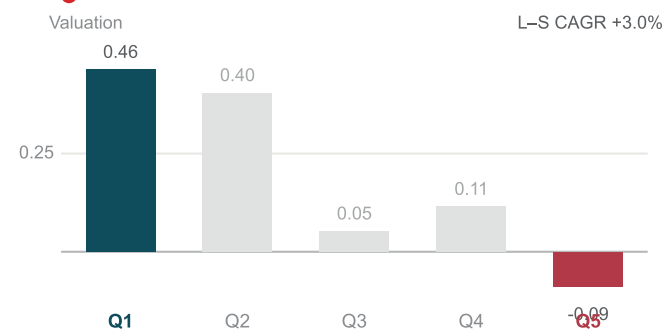
050 ● Buy-to-Sell Recommendation Ratio Less 3M EMA



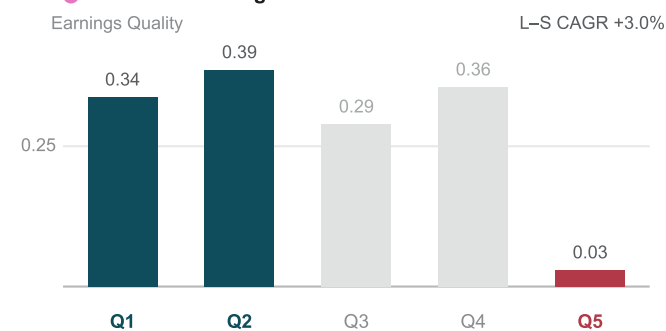
051 ● Working Capital Accruals



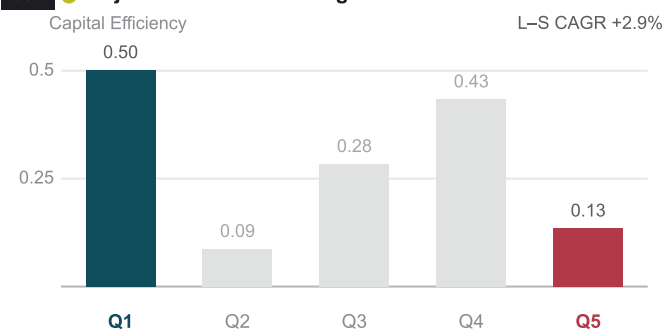
052 ● Assets to Price Ratio



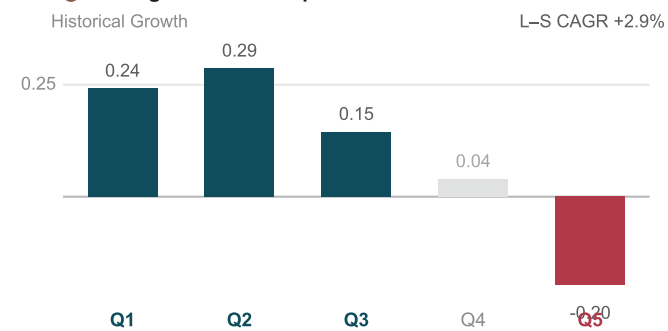
053 ● Gross Profit Margin



054 ● Adjusted Interest Coverage Ratio



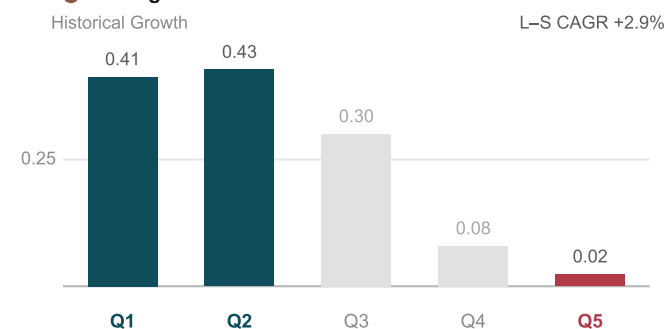
055 ● 1Y Chg in Cash Flow per Share



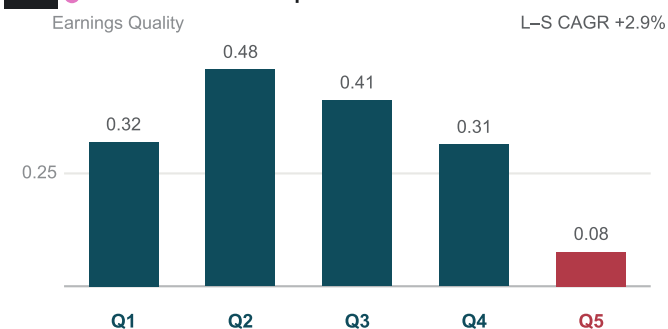
056 ● 1Y Chg in Free Cash Flow per Share



057 ● 1Y Chg in Cash Flow to Price



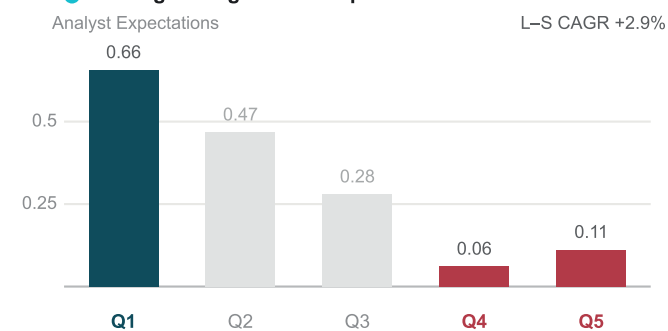
058 ● Sales to Invested Capital

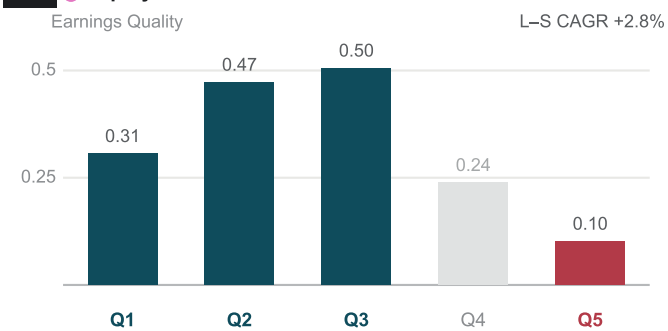
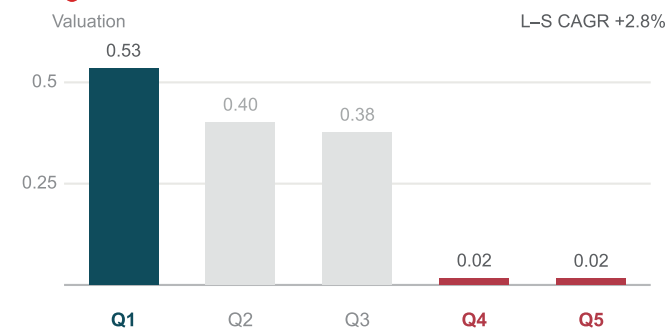
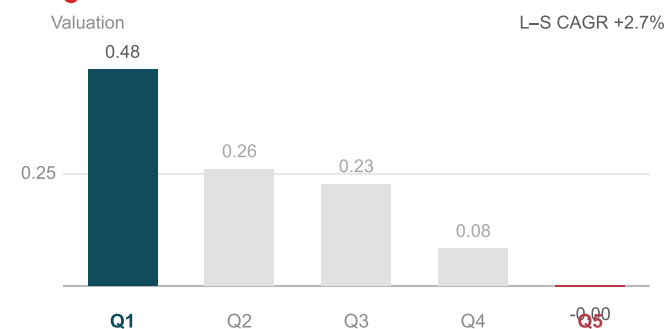
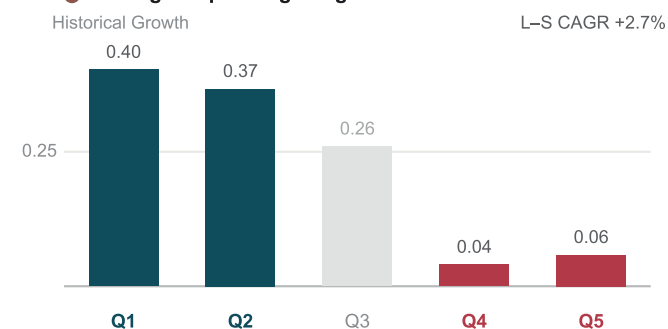
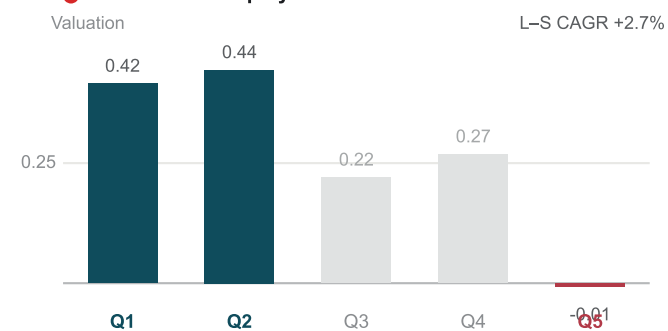
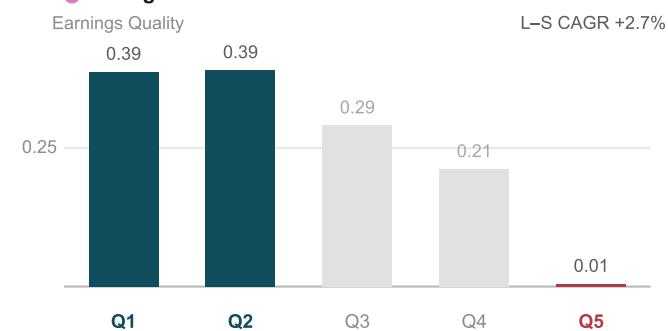


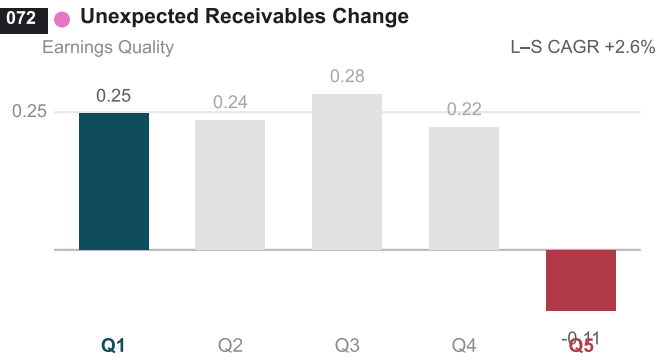
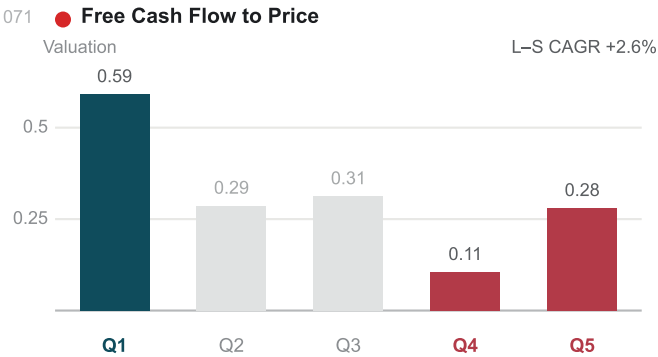
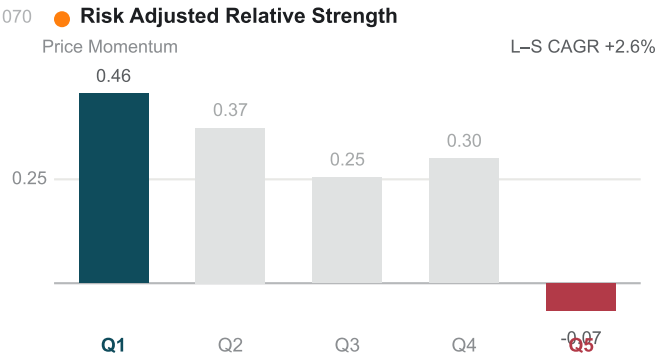
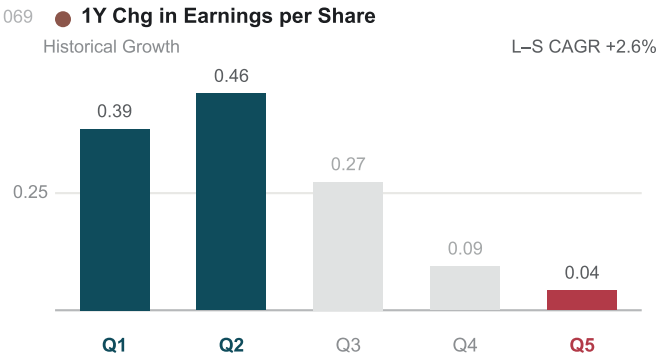
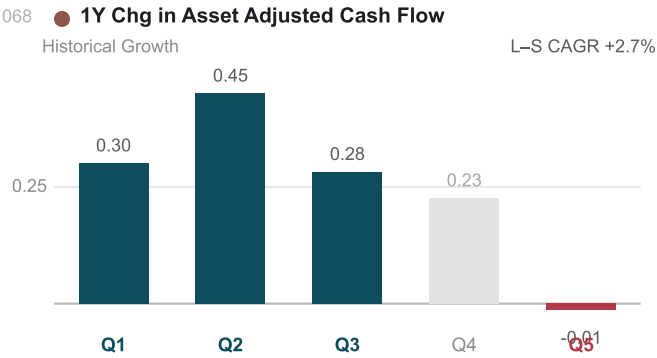
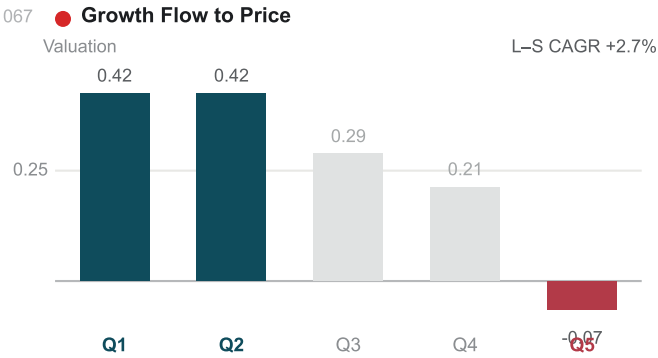
059 ● 1Y Chg in Earnings to Price

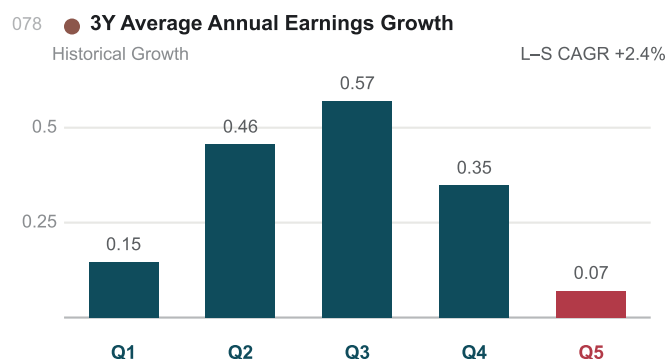
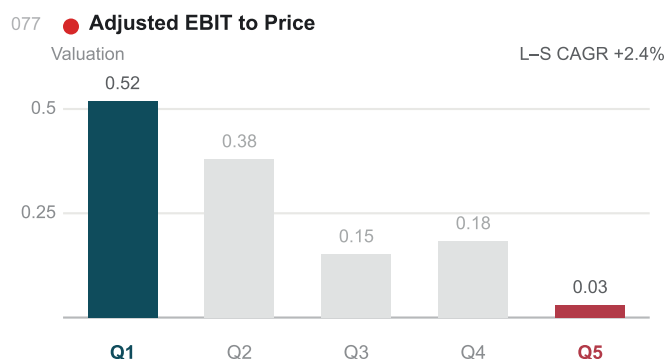
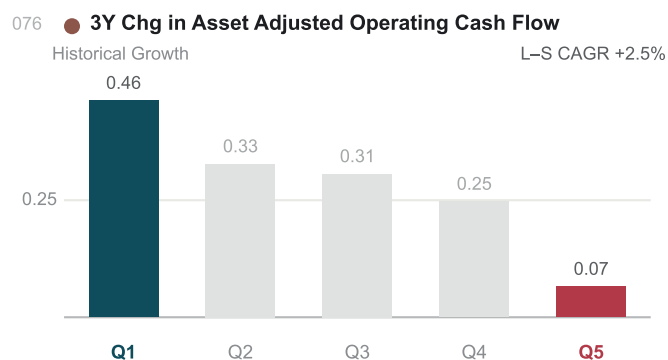
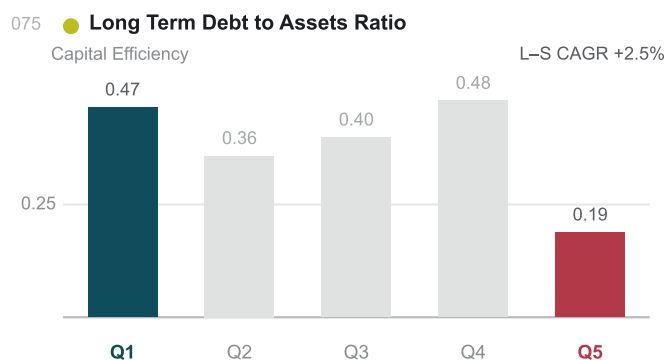
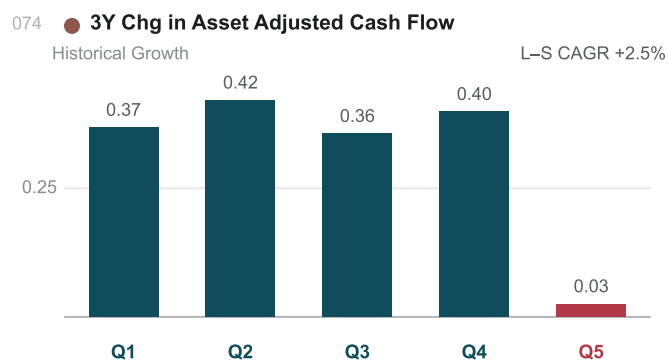
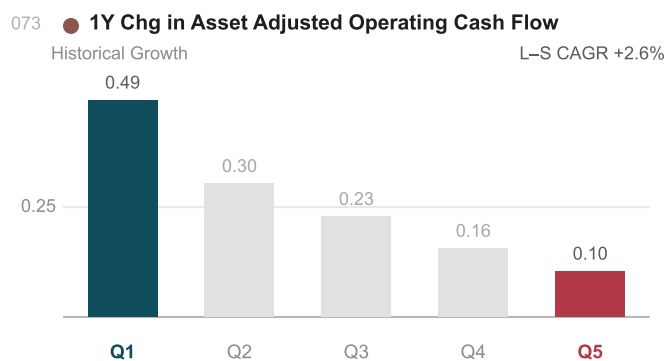


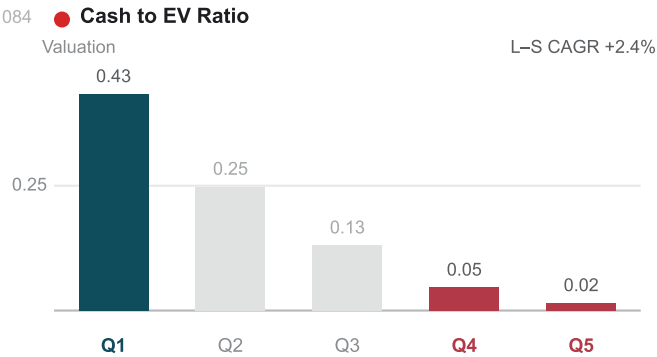
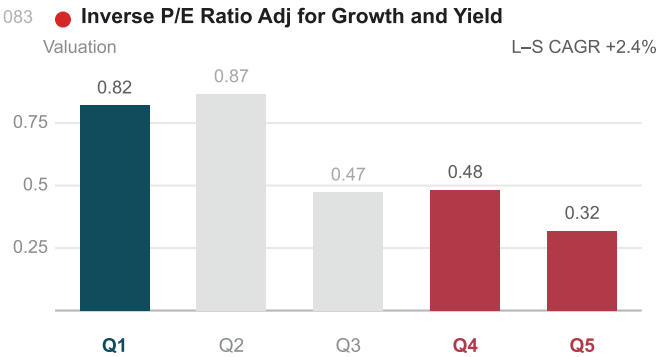
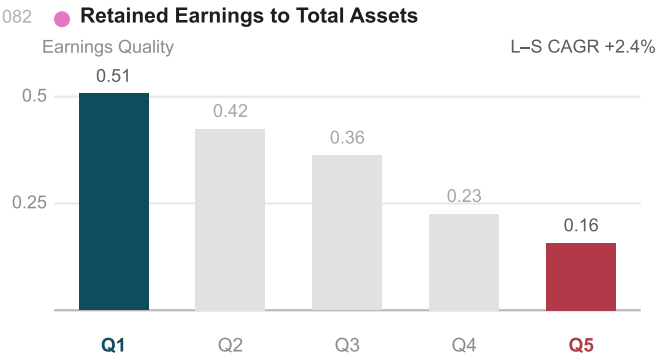
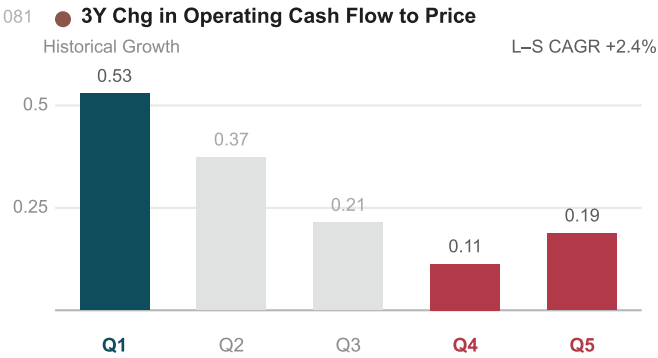
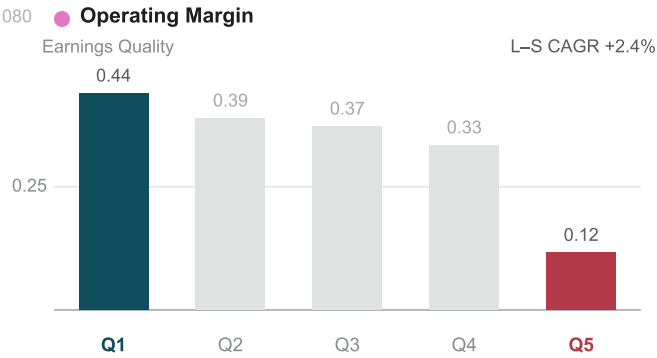
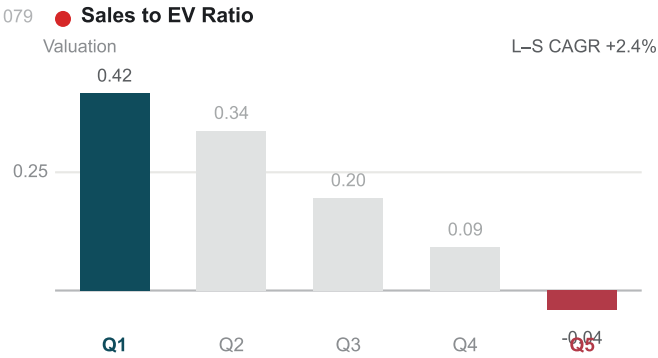
060 ● 6M Chg in Target Price Gap

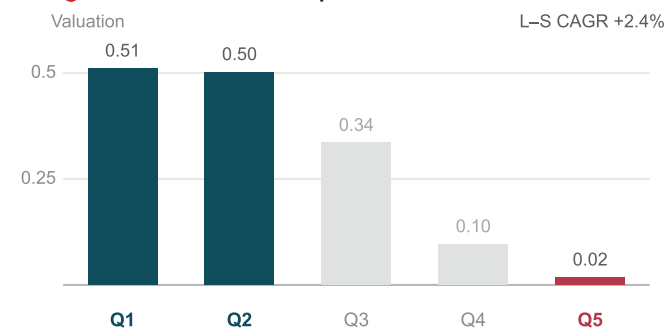
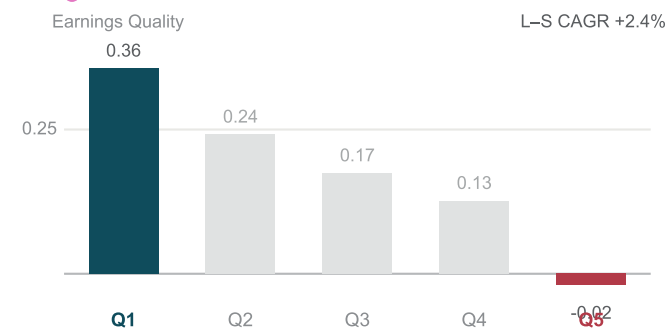
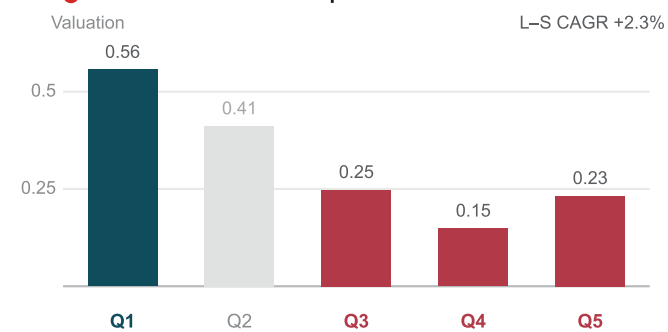
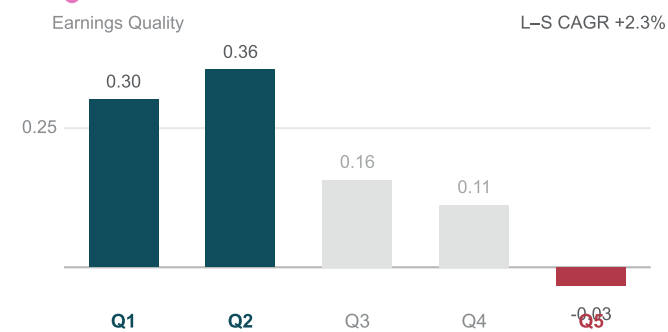
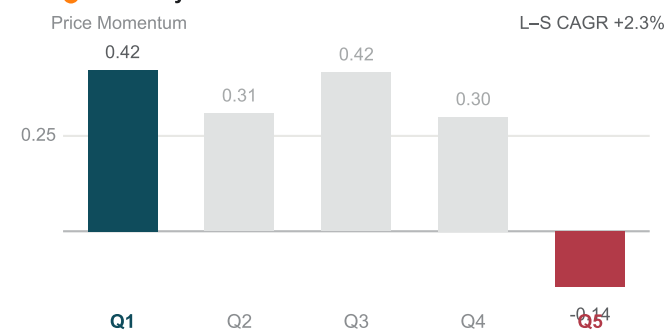
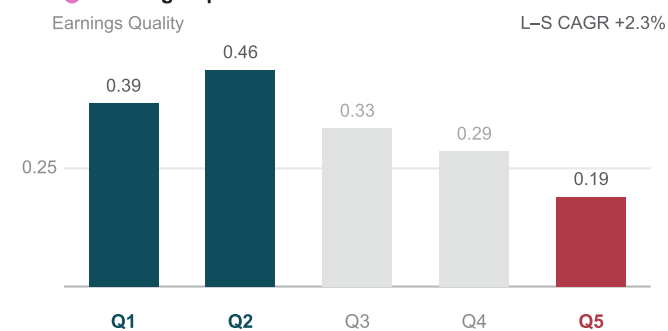


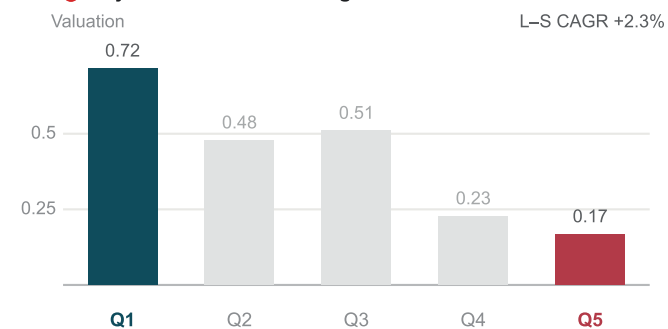
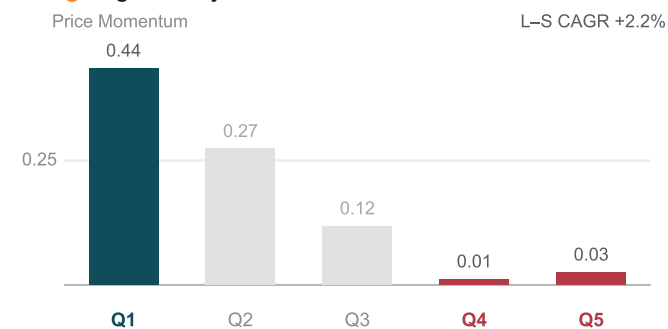
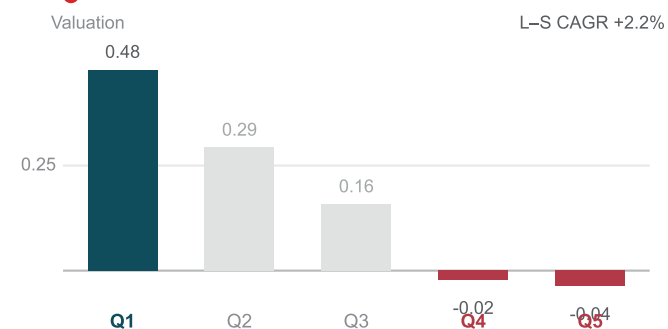
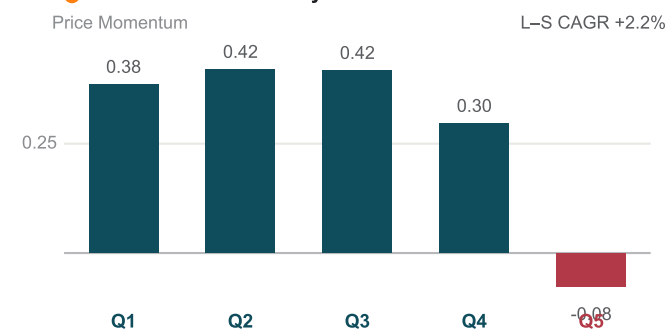
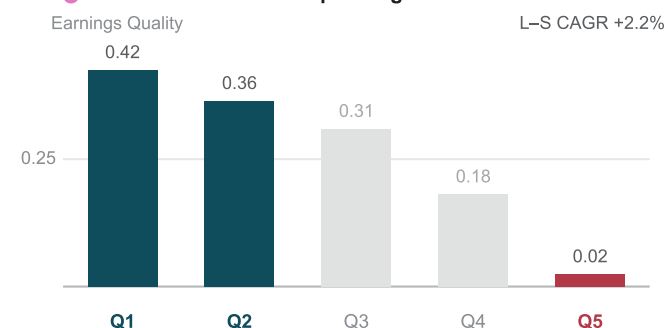
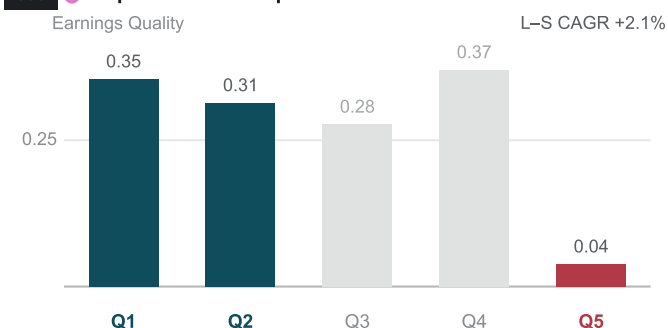
061 Equity Turnover**062 Net Cash Flow to Price****063 Cash to Price****064 1Y Chg in Operating Margin****065 Cash Flow to Equity****066 Change EPS to Sales**

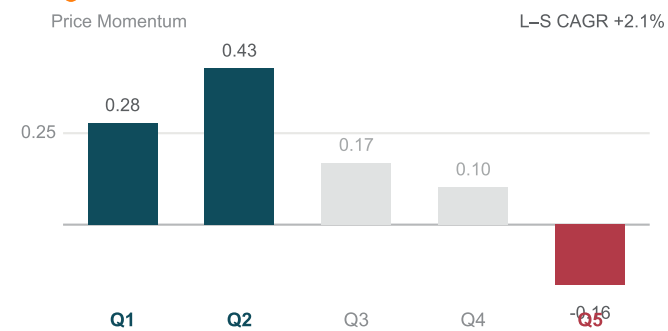
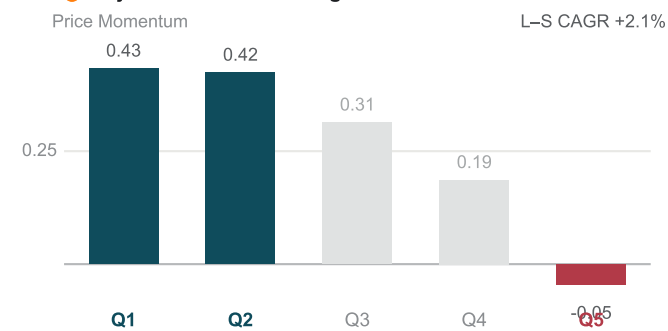
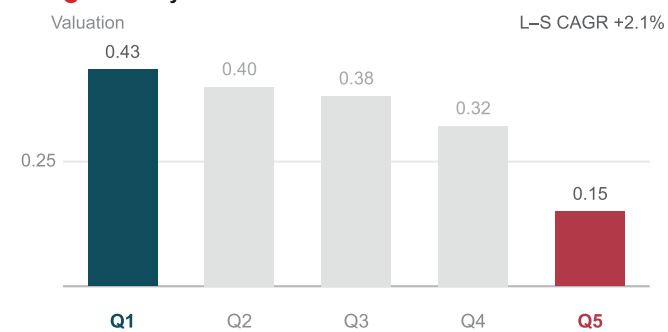
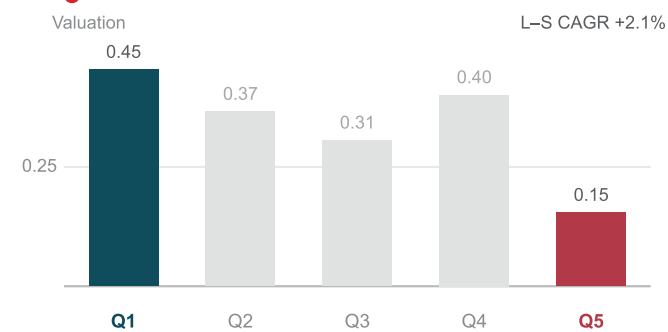
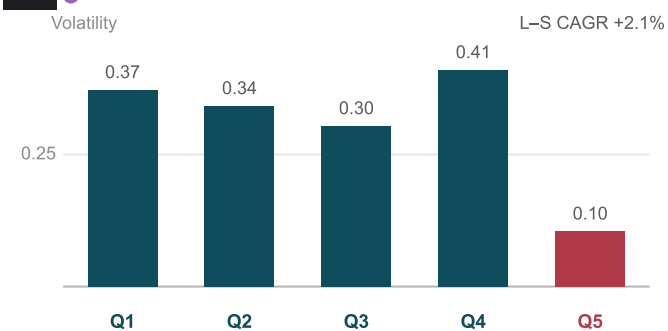
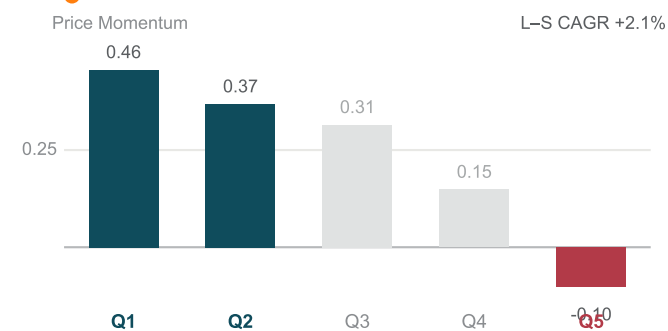


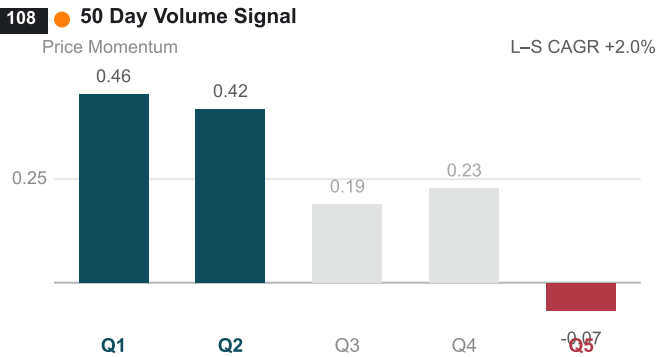
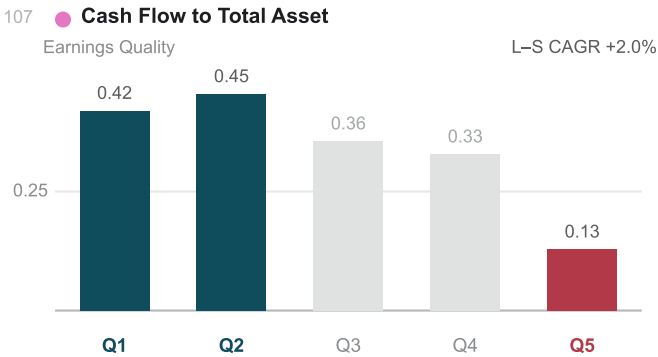
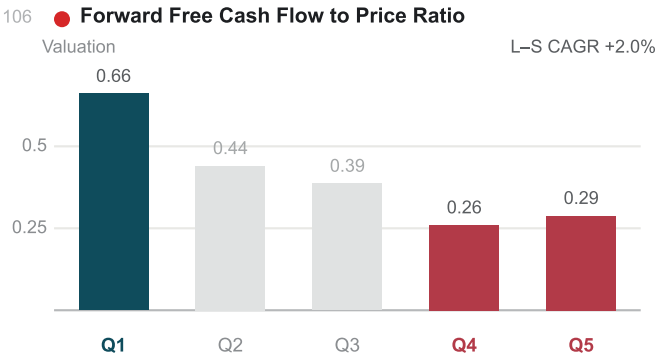
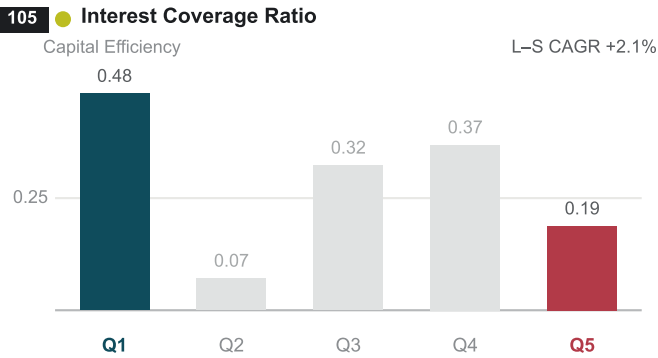
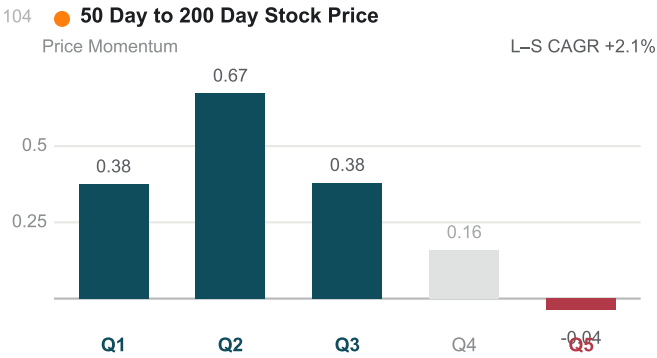


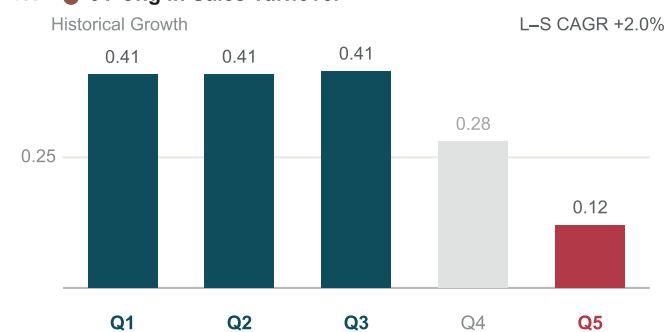
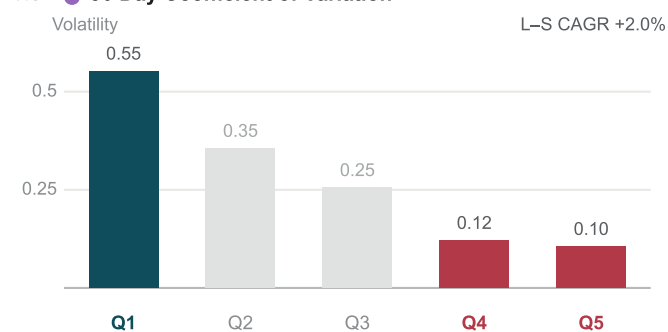
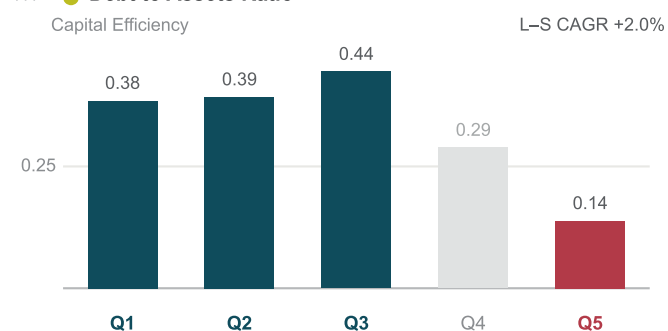
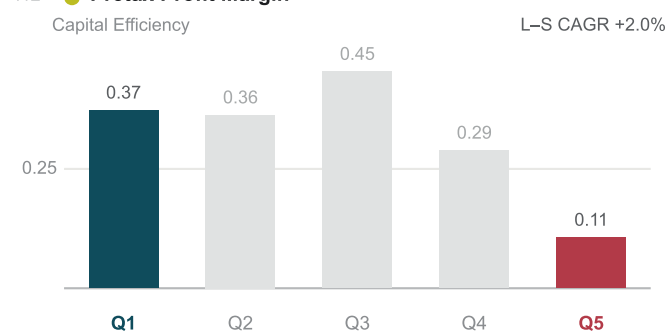
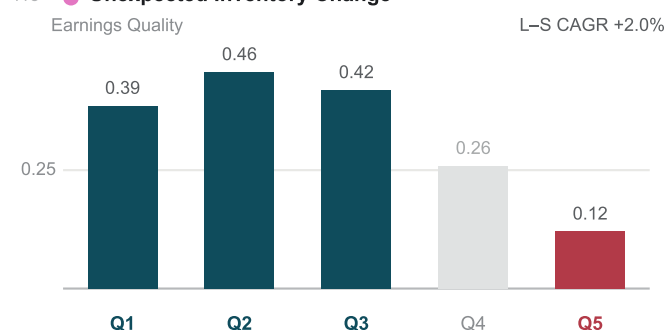
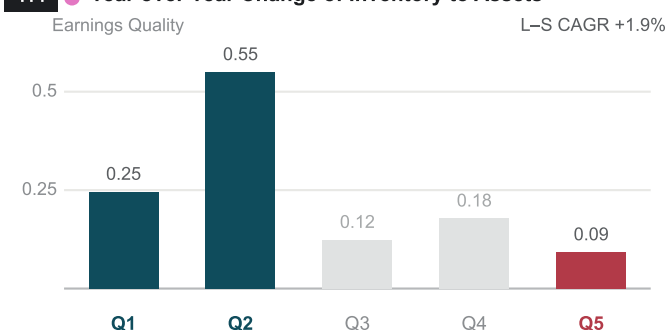


085 ● **Net Cash Flow to Enterprise Value**086 ● **Accrual Ratio - Balance Sheet**087 ● **Free Cash Flow to Enterprise Value**088 ● **Accrual Ratio - Cash Flows**089 ● **Max Daily Return in the Past 6 Months**090 ● **Working Capital to Assets Ratio**

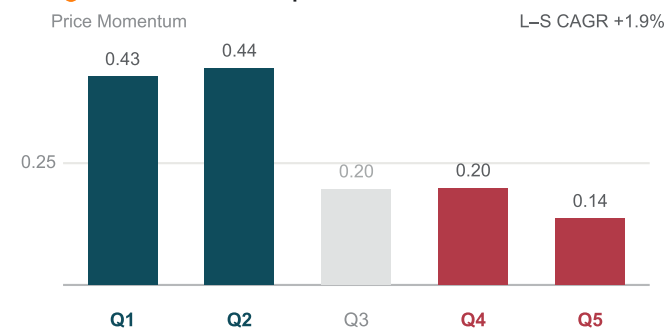
091 ● **Adjusted Forward Earnings to Price**092 ● **Log of Unadjusted Stock Price**093 ● **Book to Price**094 ● **Percent Above 260-Day Low**095 ● **Pretax Income to Net Operating Assets**096 ● **Depreciation to Capex Ratio**

097 ● **15/36 Week Price Ratio**098 ● **Adjusted 6 Month Average Share Turn Over**099 ● **Solvency Ratio**100 ● **Cash to Total Assets**101 ● **60M CAPM Beta**102 ● **12M - 1M Price Momentum**



109 ● **3Y Chg in Sales Turnover**110 ● **90 Day Coefficient of Variation**111 ● **Debt to Assets Ratio**112 ● **Pretax Profit Margin**113 ● **Unexpected Inventory Change**114 ● **Year over Year Change of Inventory to Assets**

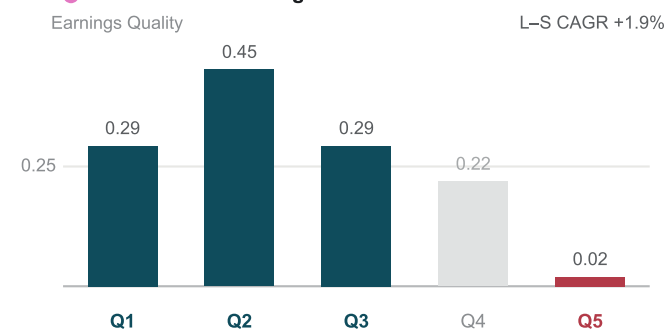
115 ● Institution Ownership Level



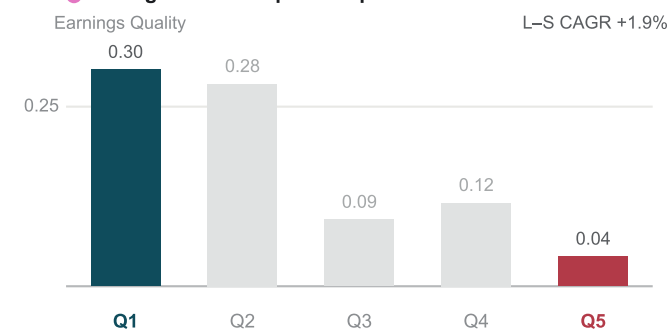
116 ● Dividend Growth



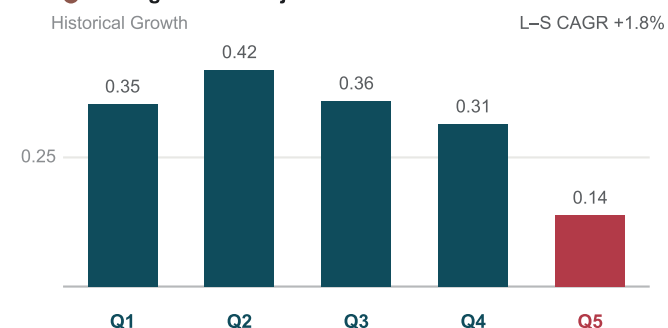
117 ● Year over Year Change of EPS to Sales



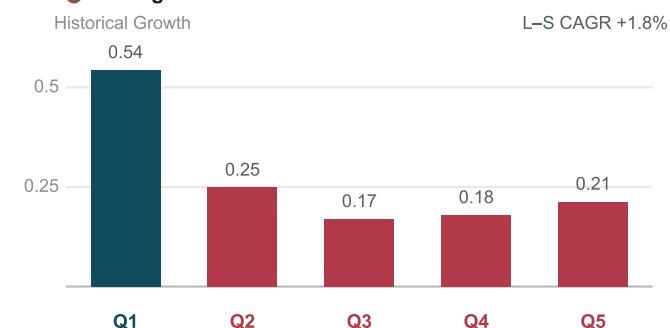
118 ● Change in TTM Depr. to Capex



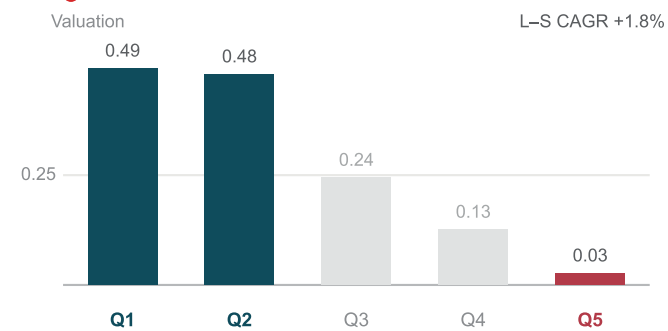
119 ● 3Y Chg in Asset Adjusted EPS



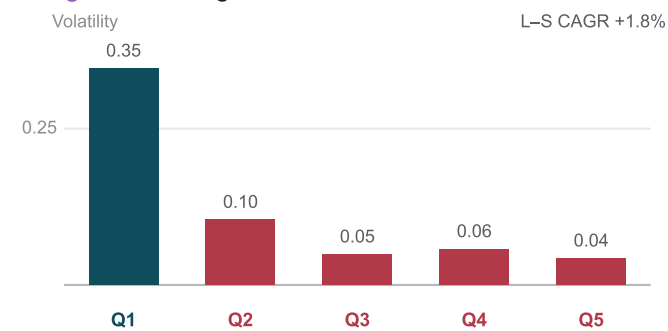
120 ● 3Y Chg in Cash Flow to Price



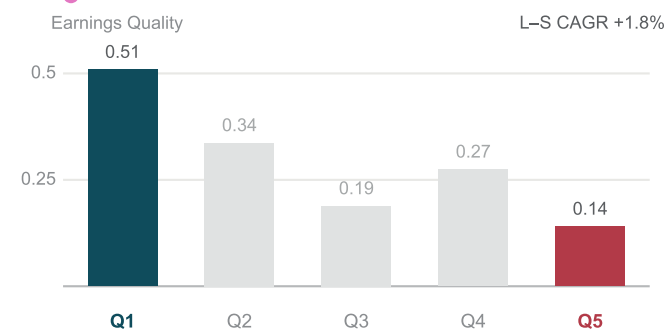
121 ● Pretax Income Per Share to Price



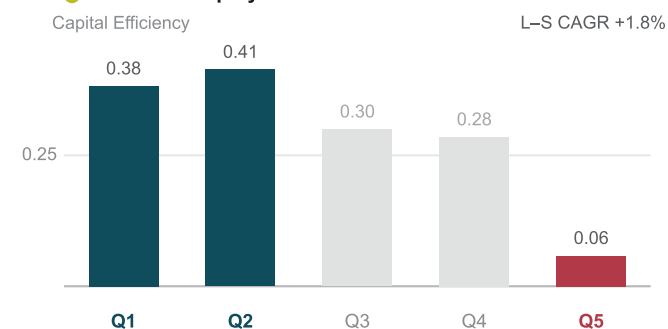
122 ● Book Leverage



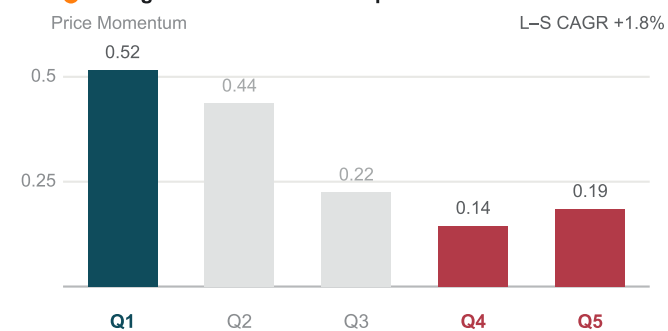
123 ● Cash Burn Rate



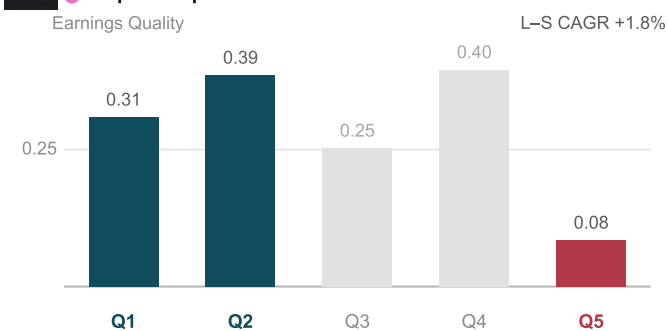
124 ● Return on Equity

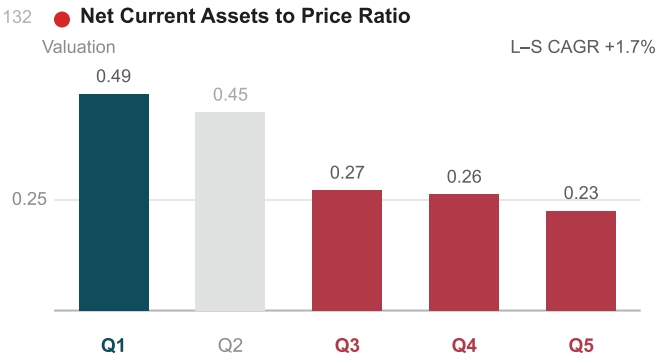
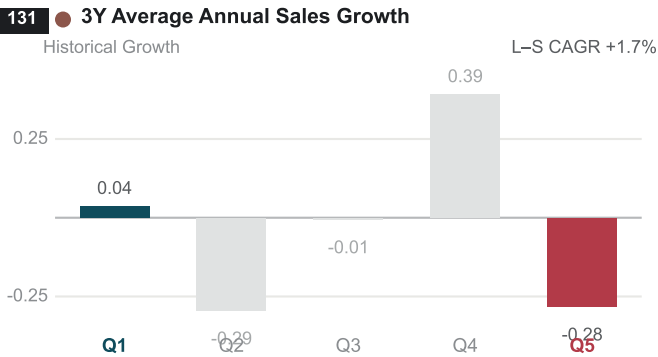
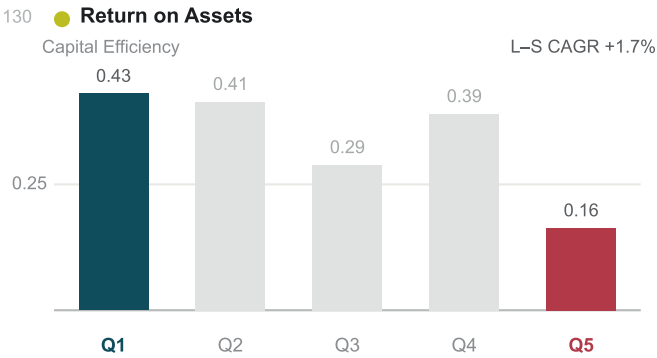
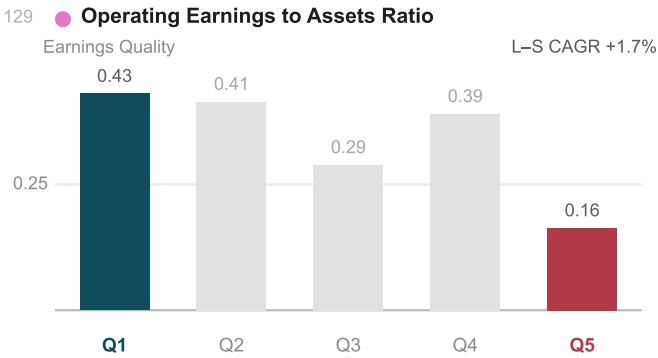
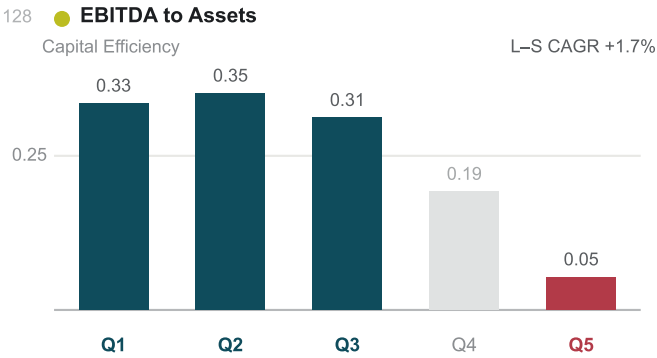
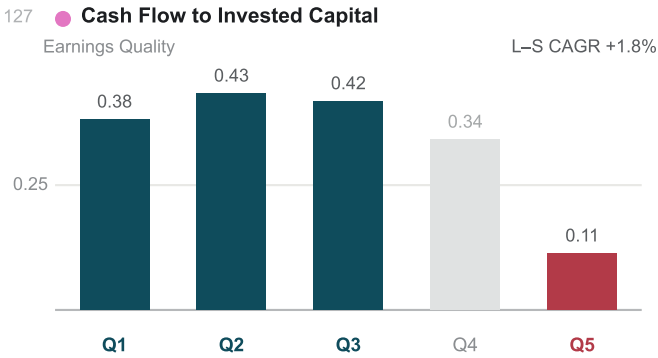


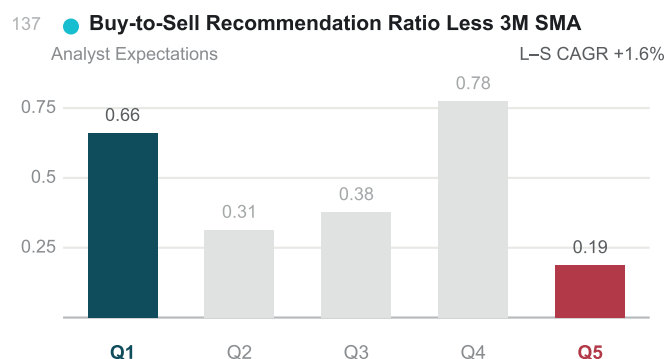
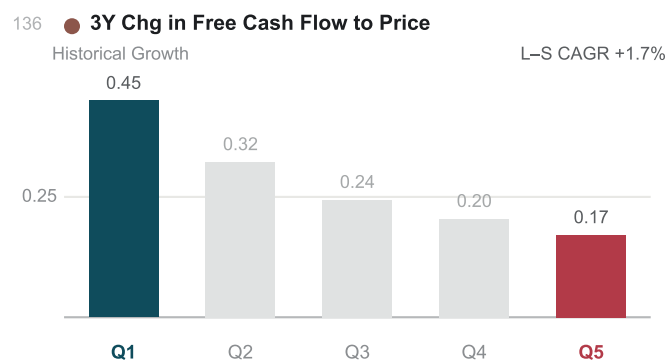
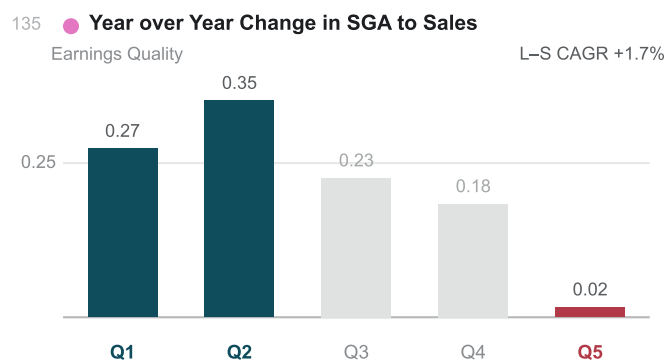
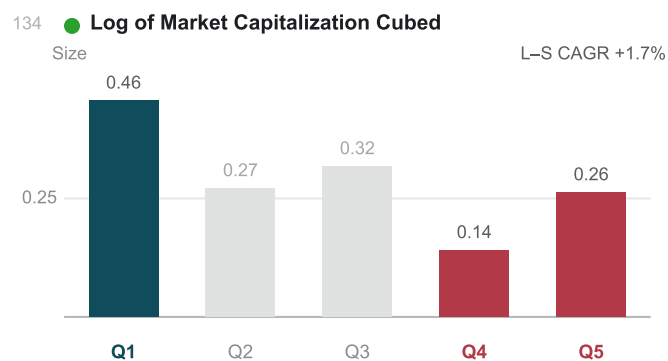
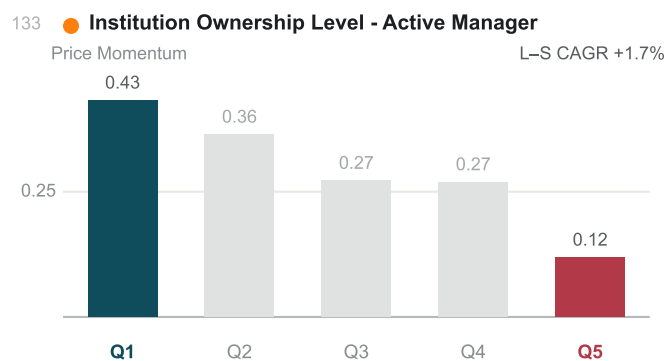
125 ● Foreign Institution Ownership



126 ● Capital Expenditure to Sales



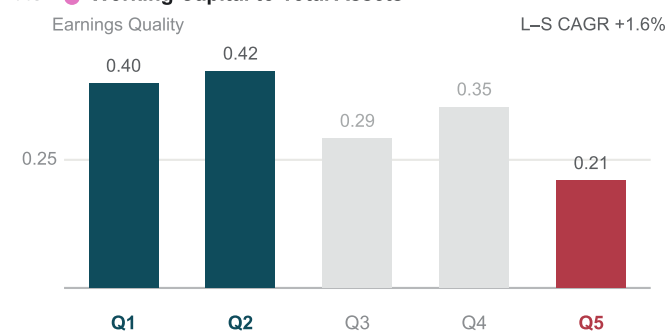




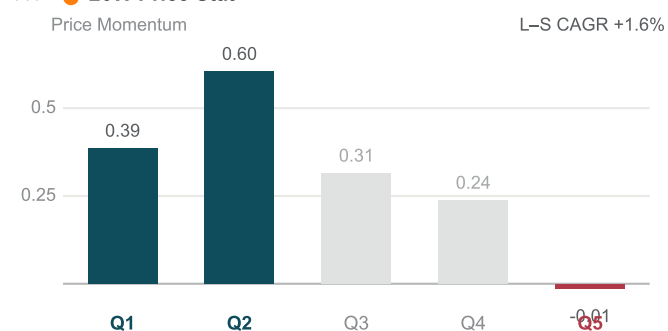
139 ● 3Y Chg in Asset Adjusted Free Cash Flow



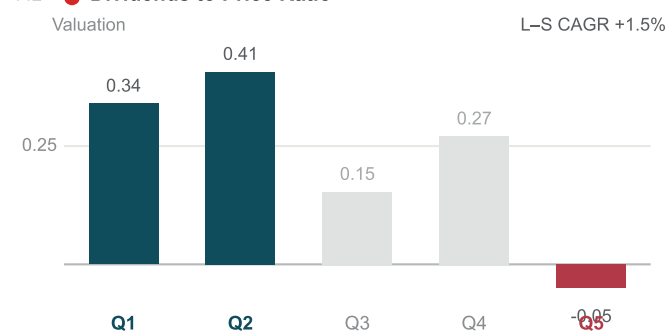
140 ● Working Capital to Total Assets



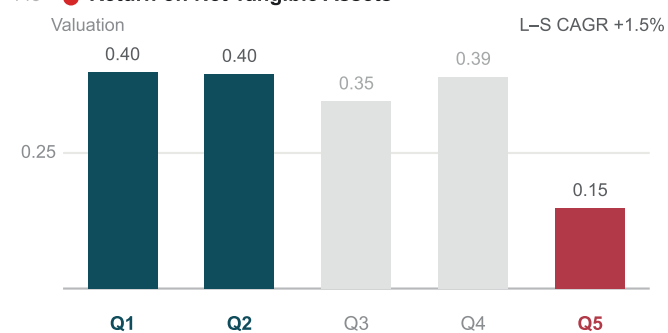
141 ● 26W Price Stat



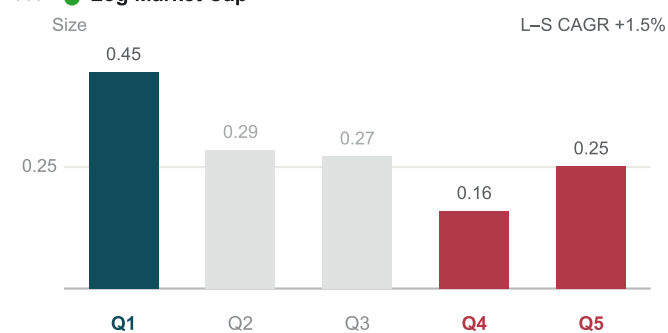
142 ● Dividends to Price Ratio



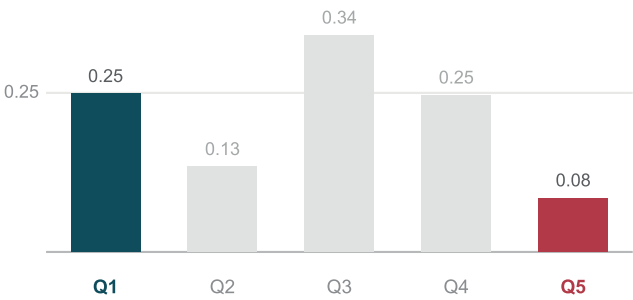
143 ● Return on Net Tangible Assets



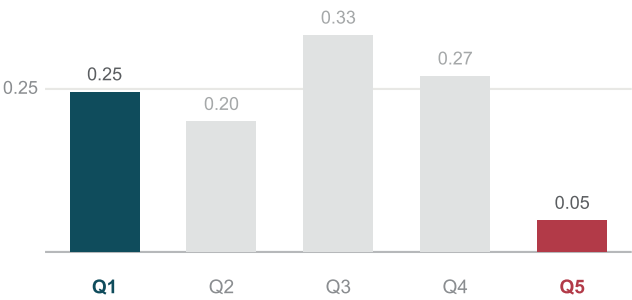
144 ● Log Market Cap



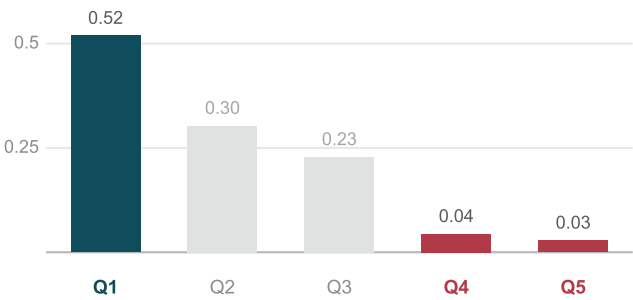
145 ● **Trailing 12-Month Receivables and Inventories/Trailing 12-M...**
Historical Growth L-S CAGR +1.5%



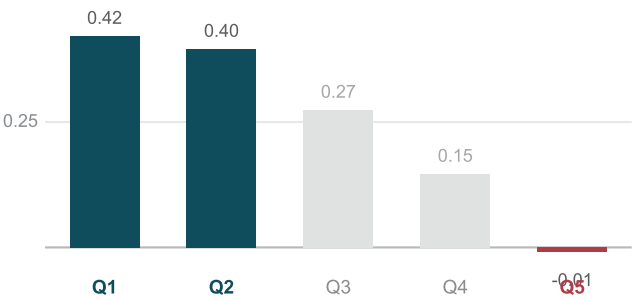
146 ● **Working Capital Turnover Ratio**
Capital Efficiency L-S CAGR +1.5%



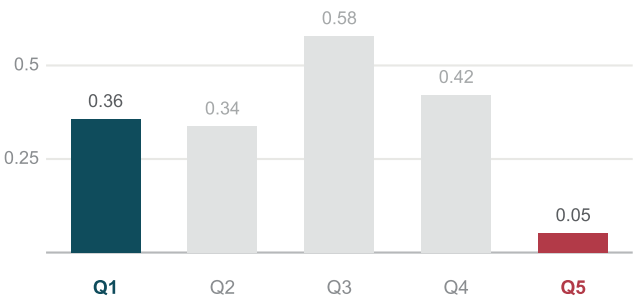
147 ● **EBITDA to Price**
Valuation L-S CAGR +1.4%



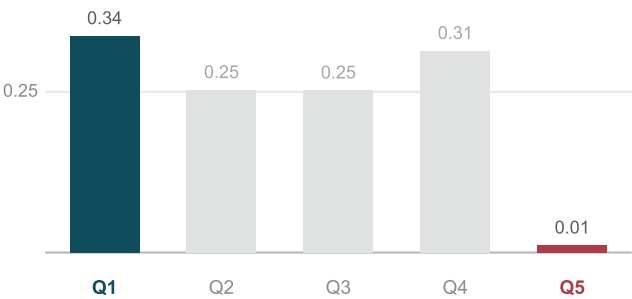
148 ● **Earnings to Price**
Valuation L-S CAGR +1.4%



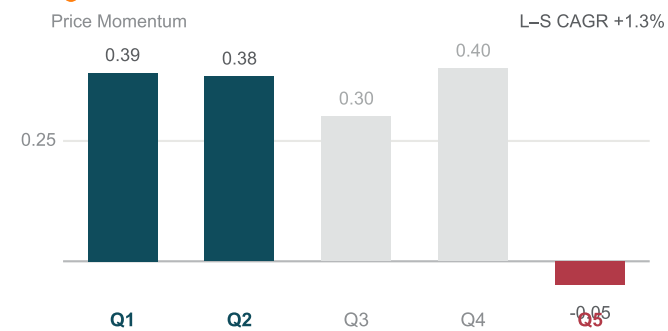
149 ● **3Y Chg in Cash Flow per Share**
Historical Growth L-S CAGR +1.4%



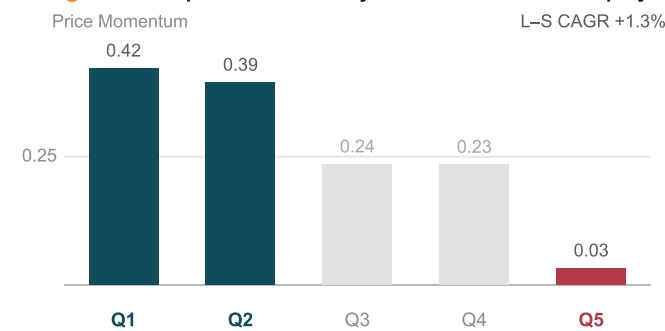
150 ● **Asset Growth - 1 Year**
Capital Efficiency L-S CAGR +1.4%



151 ● 6 Months Price Reversal



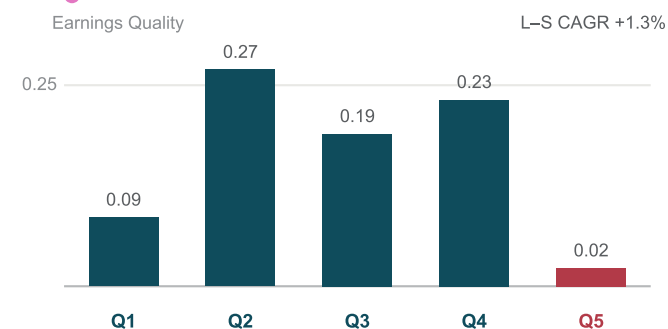
152 ● Ownership Breadth Stability of Institution Ownership by Buyers



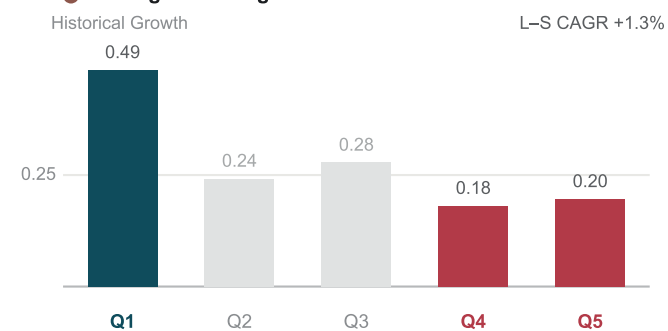
153 ● Forward Earnings to Price



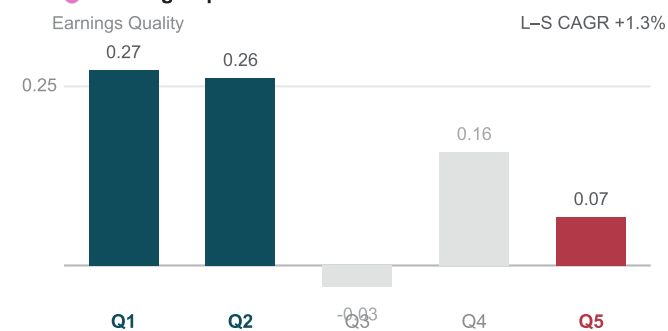
154 ● Cash Ratio



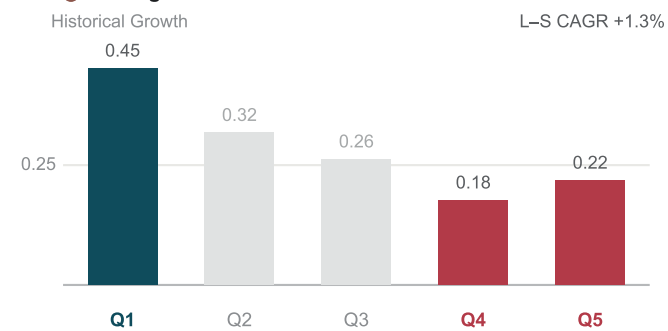
155 ● 3Y Chg in Earnings to Price



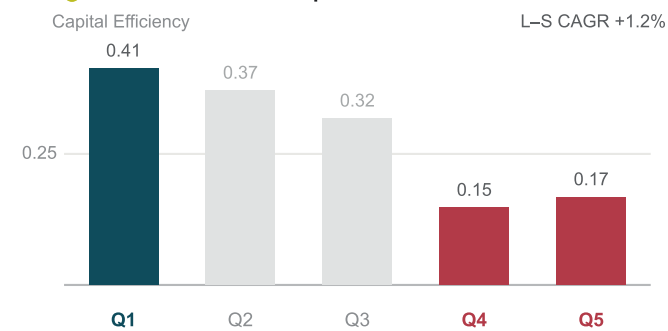
156 ● Working Capital to Sales



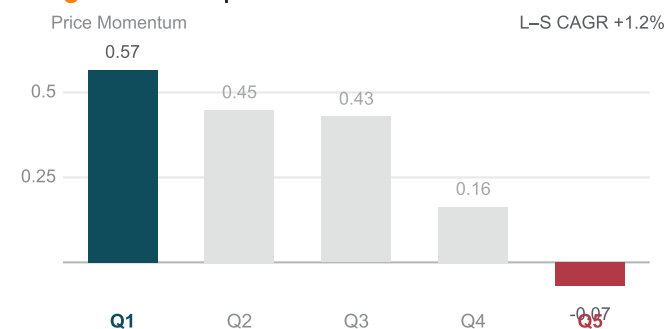
157 ● 3Y Chg in Sales to Price



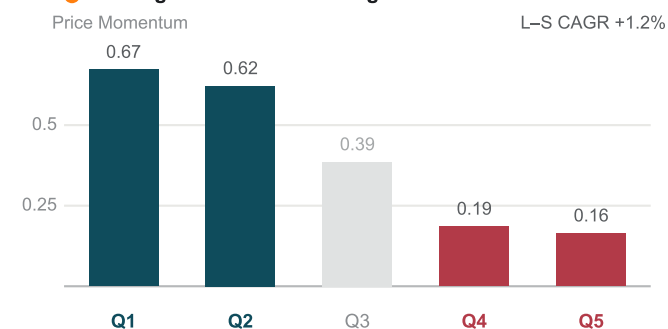
158 ● Return on Invested Capital



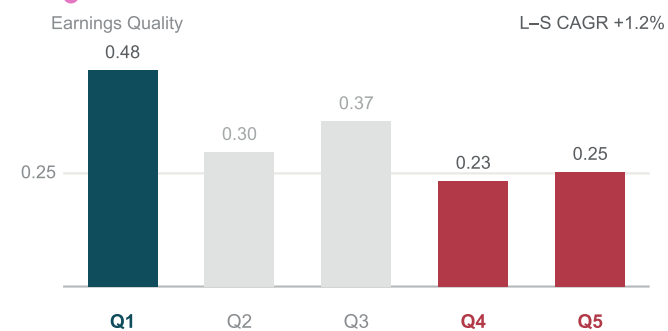
159 ● 60M CAPM Alpha



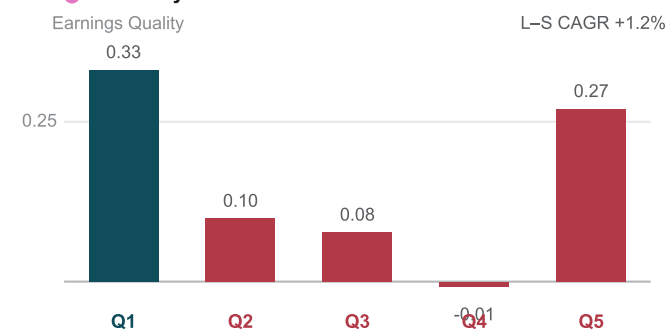
160 ● Closing Price to 52 Week High

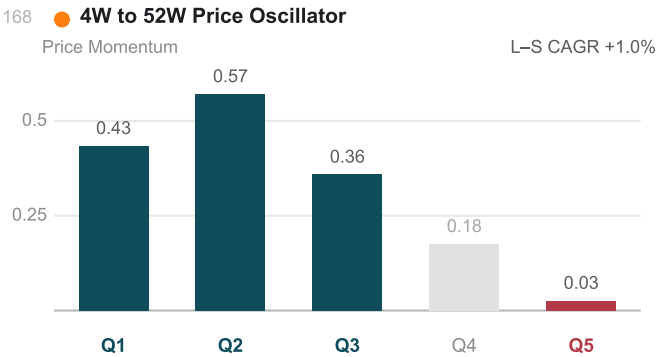
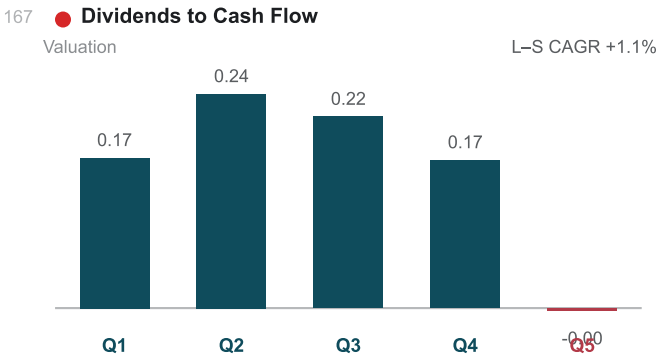
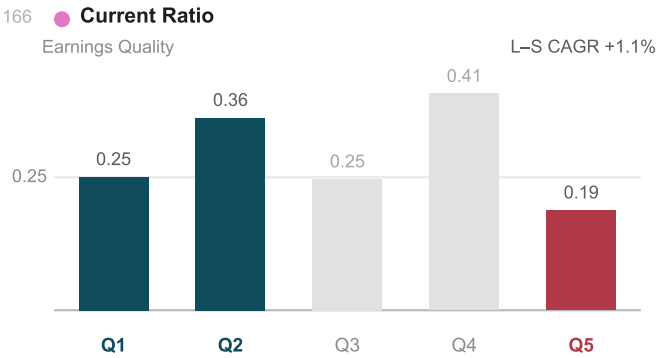
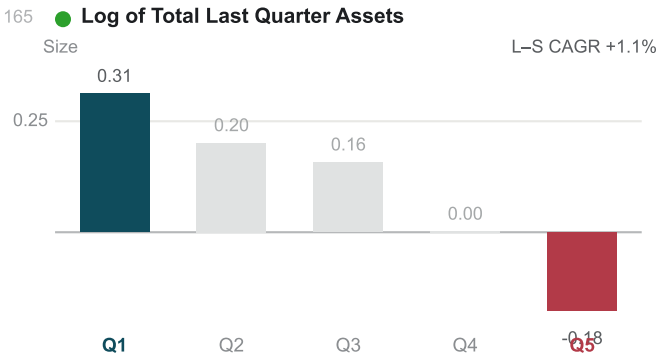
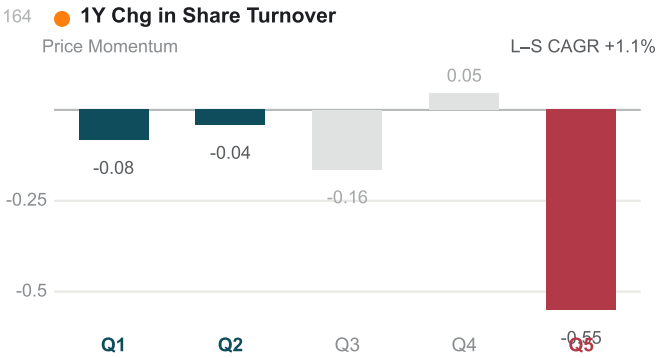
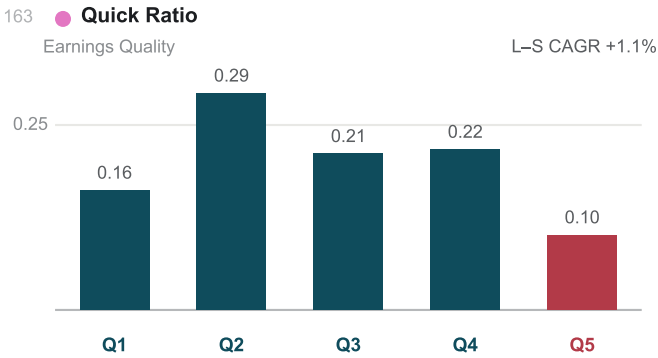


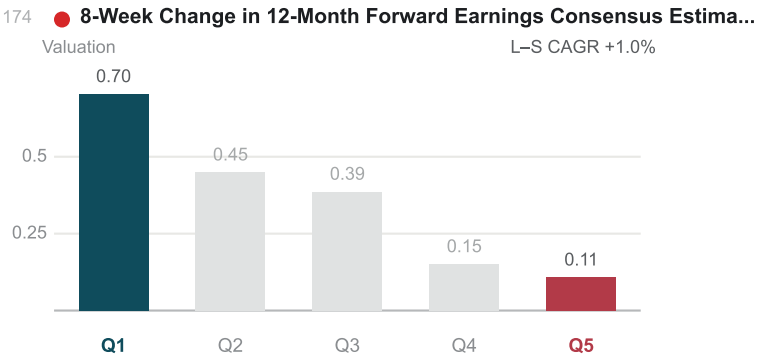
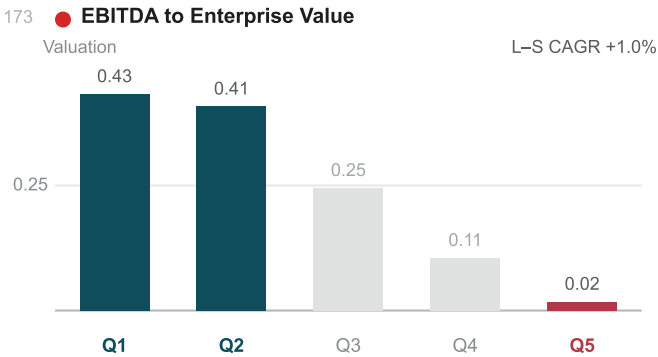
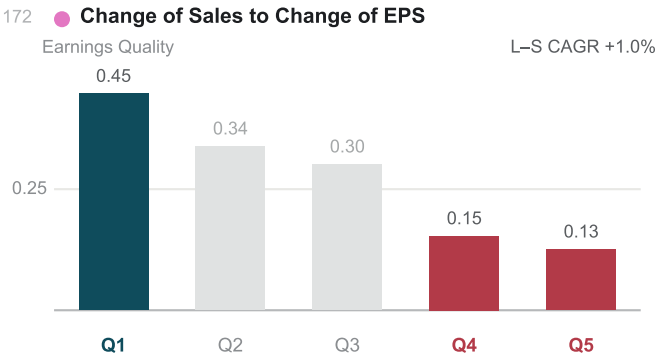
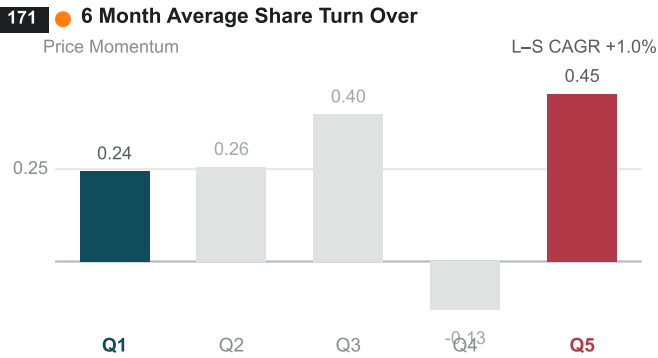
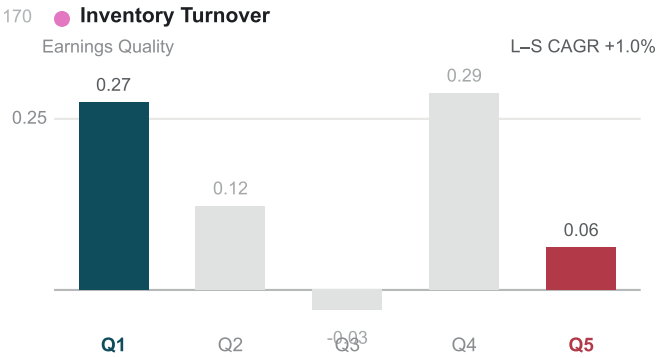
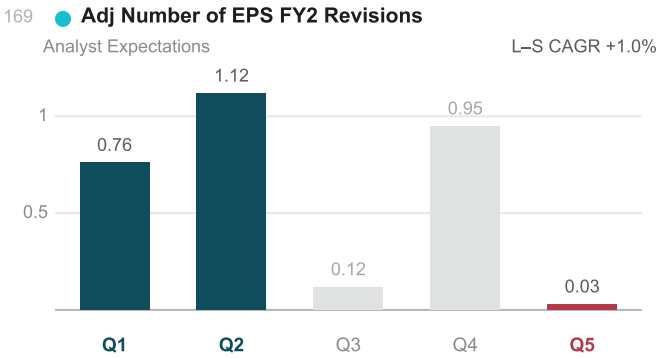
161 ● Cash to Sales

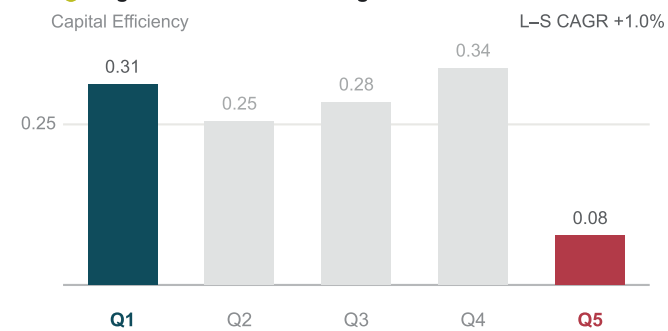
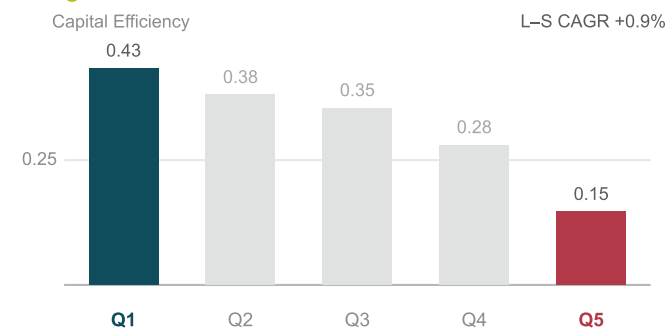
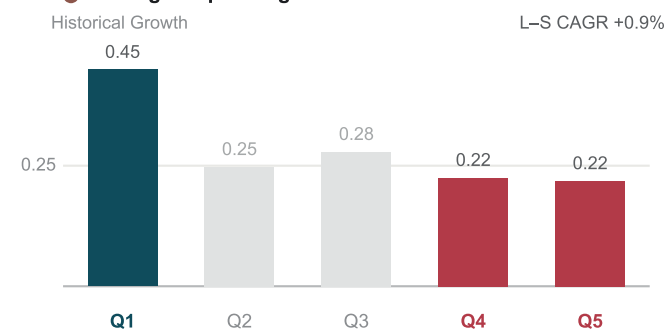
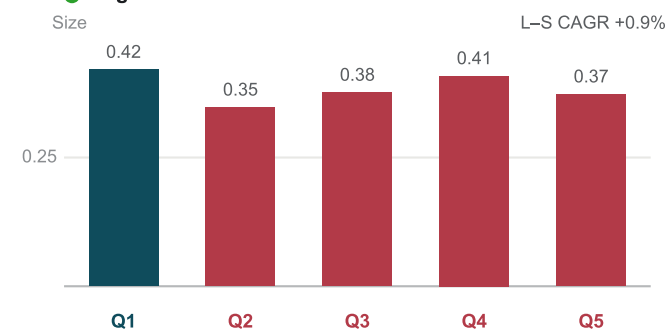
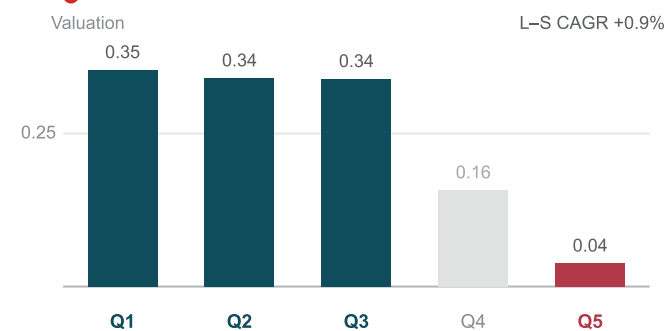
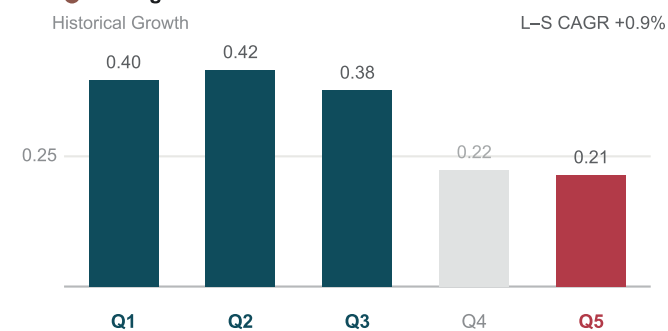


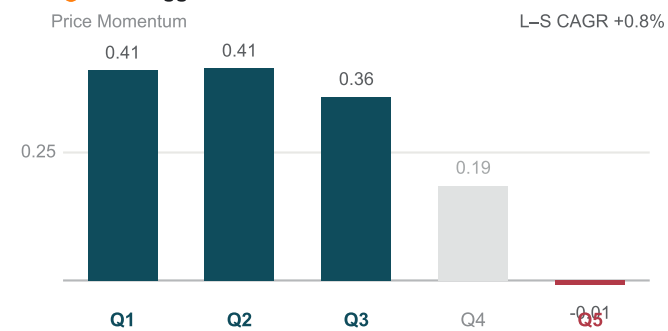
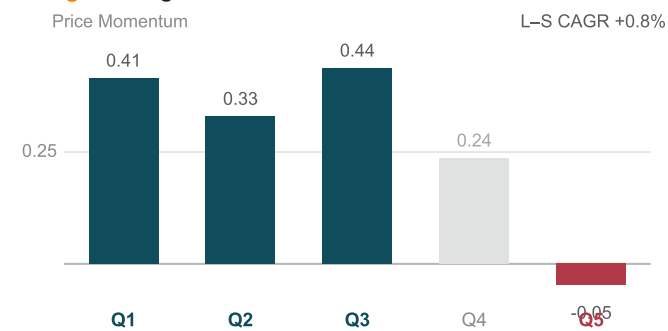
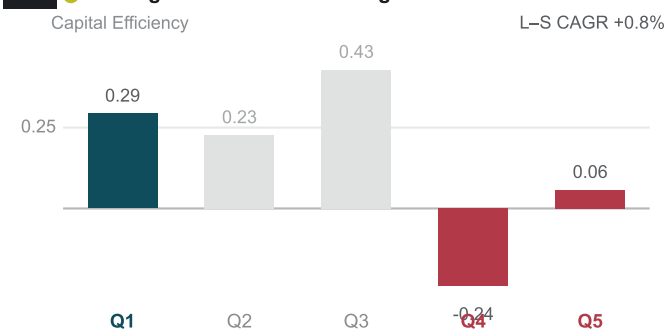
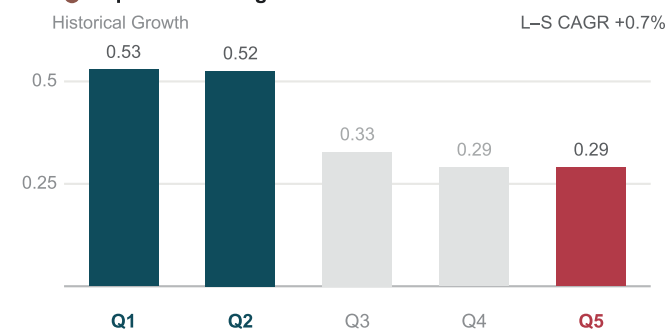
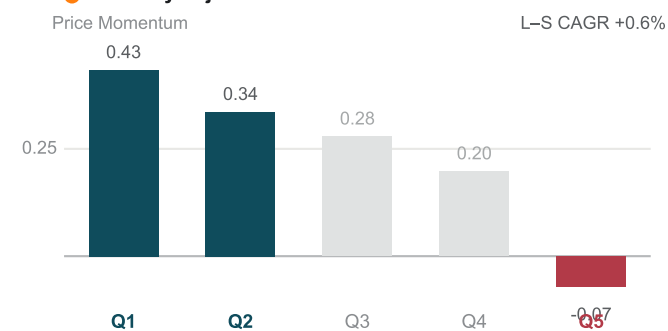
162 ● Inventory to Assets Ratio

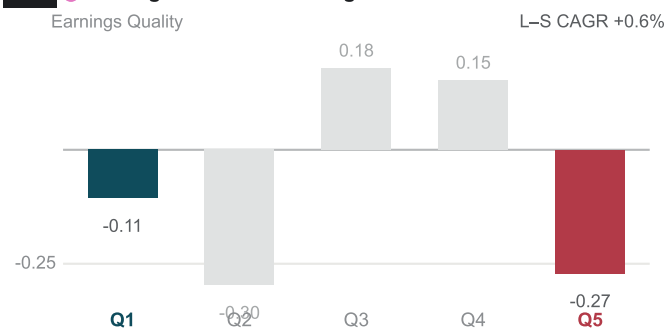
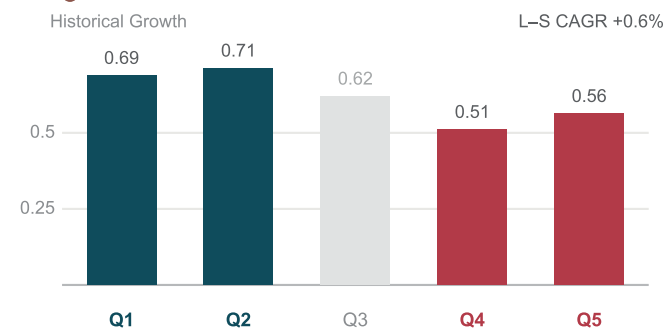
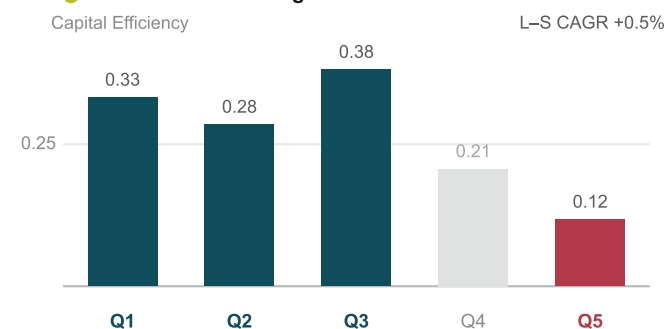
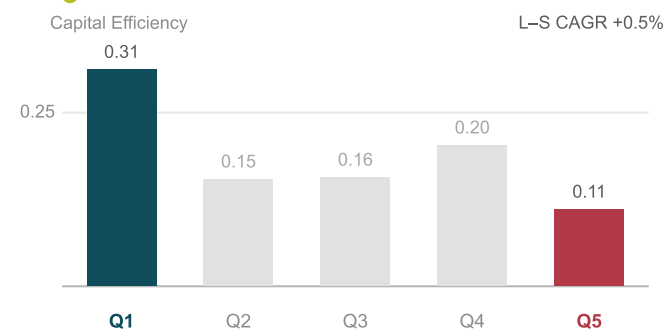
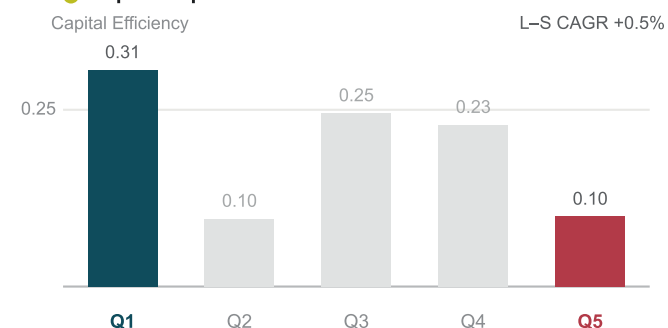
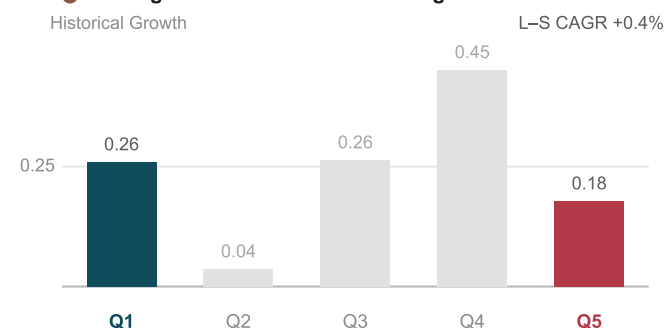


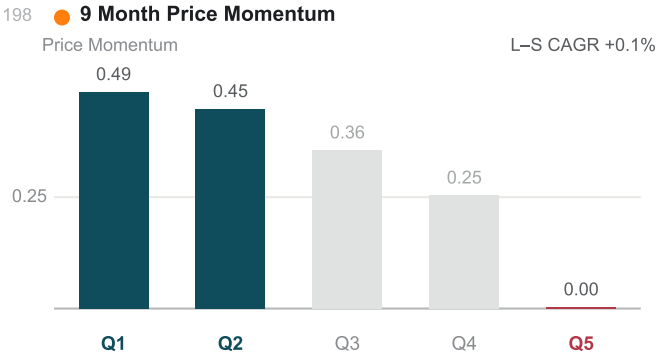
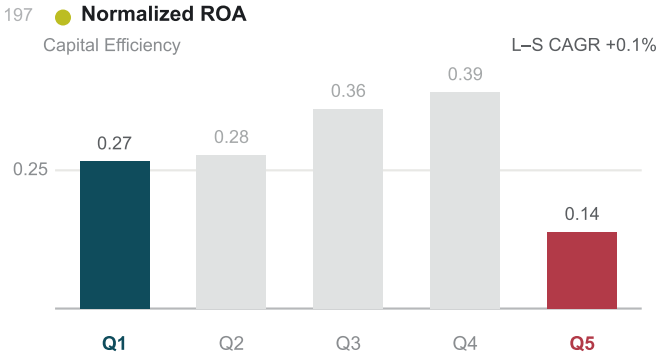
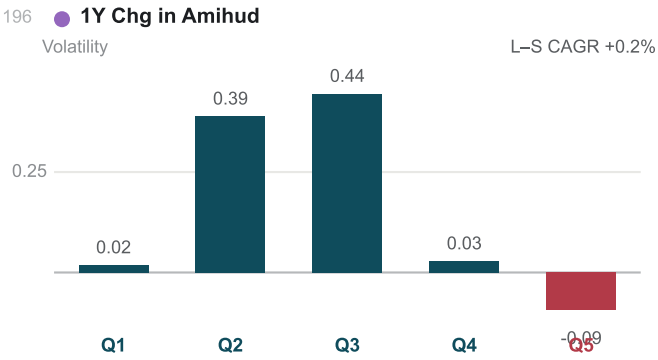
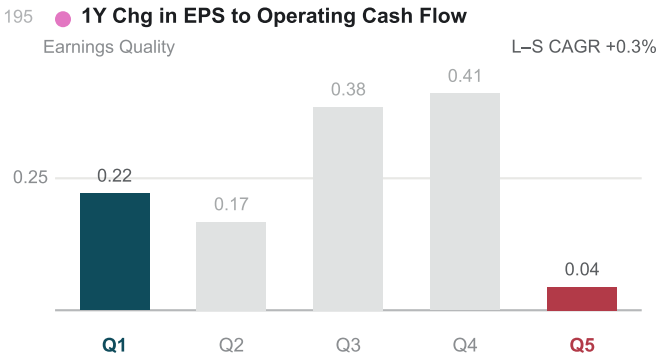
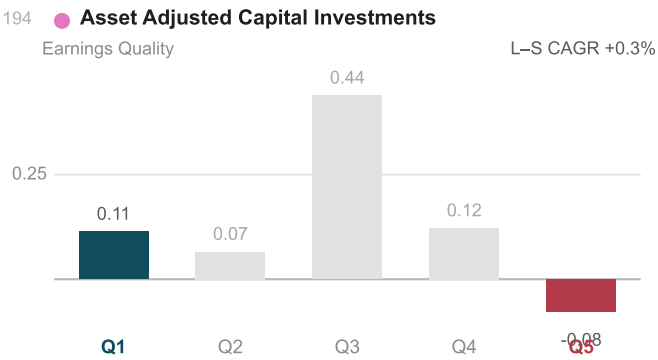
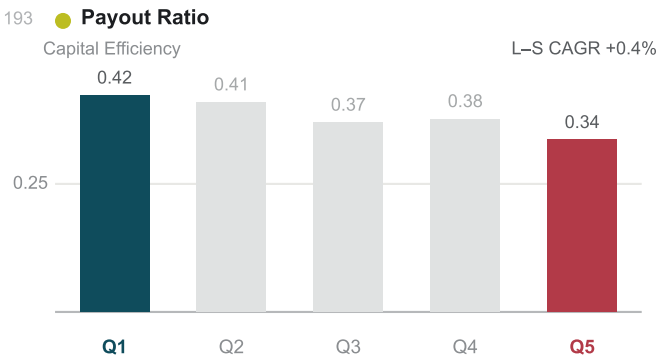




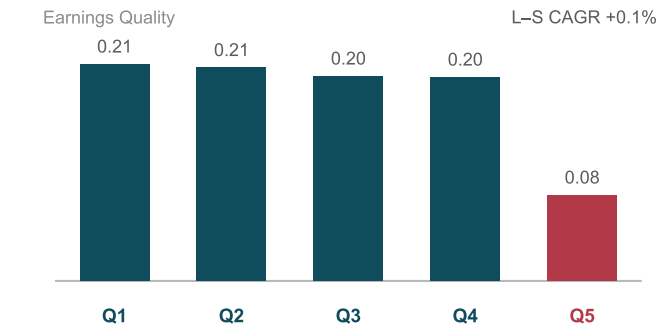
175 ● **Degree of Financial Leverage**176 ● **EBIT to Assets**177 ● **1Y Chg in Operating Cash Flow to Price**178 ● **Log TTM Sales**179 ● **Normalized EP**180 ● **1Y Chg in Free Cash Flow to Price**

181 ● **39W Lagged Return**182 ● **52W High Low**183 ● **1Y Chg in Shares Outstanding**184 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**185 ● **Cash Conversion Cycle**186 ● **Volatility adjusted 12M return**

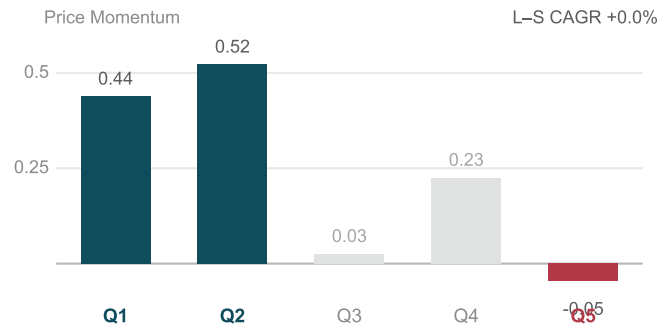
187 1Y Chg in Sales to Earnings**188 Sustainable Growth Rate****189 Year over Year Change of Total Debt****190 Normalized ROE****191 Capital Expenditure to Total Assets****192 Trailing 12-Month Overhead/Trailing 12-Month Sales**



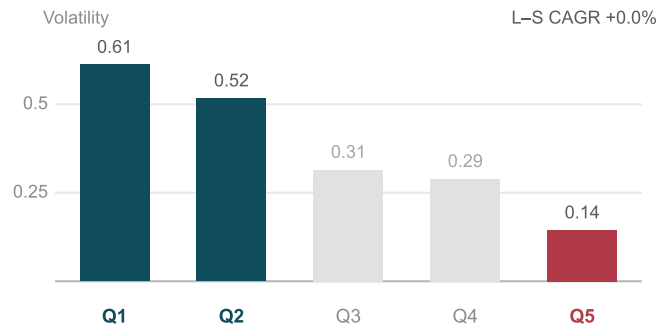
199 Balance Sheet Accruals



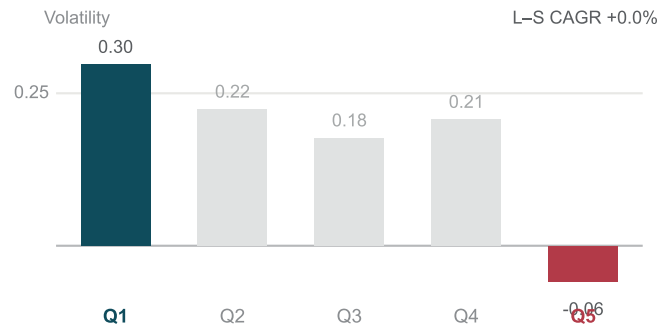
200 6M Chg in 12M CAPM Alpha



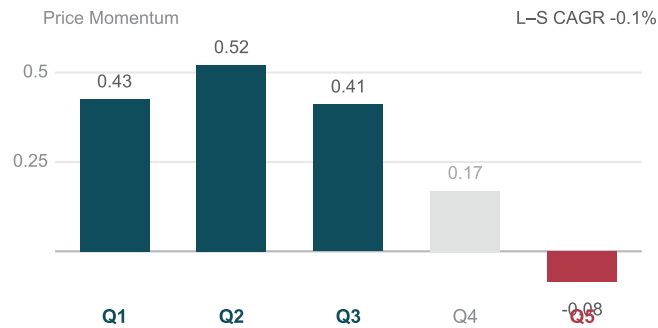
201 1M Realized Price Volatility



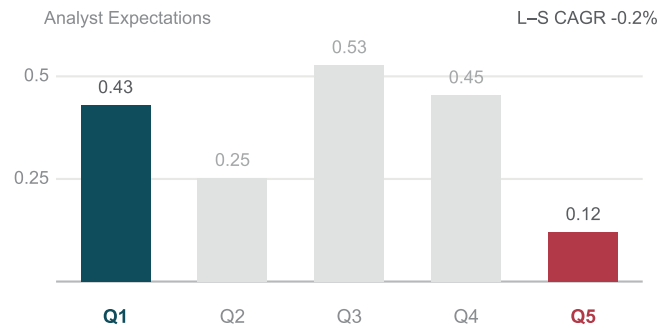
202 130 Day Minimum Return



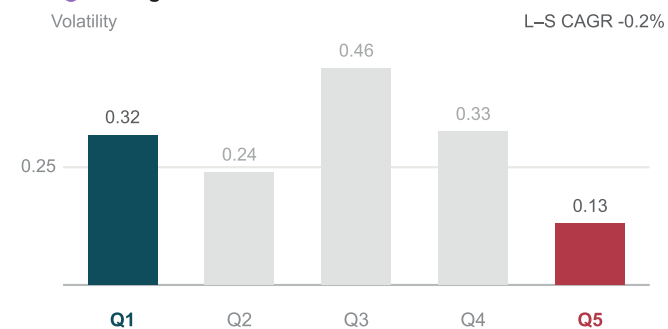
203 5 Day Price Reversal



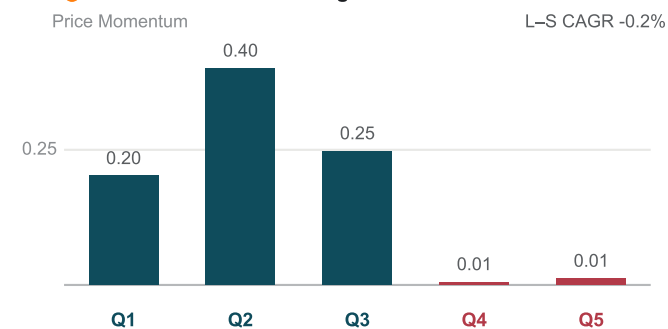
204 Analyst Dispersion for FY2 EPS



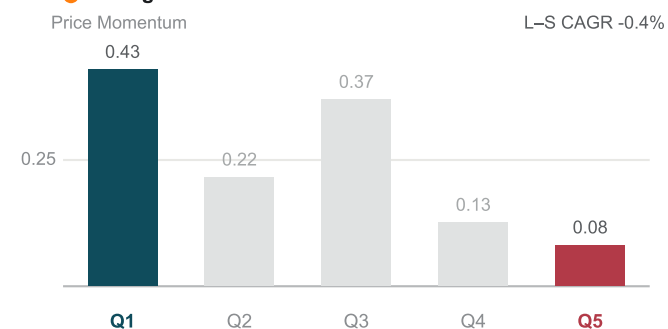
205 ● 1Y Chg in Debt to Assets



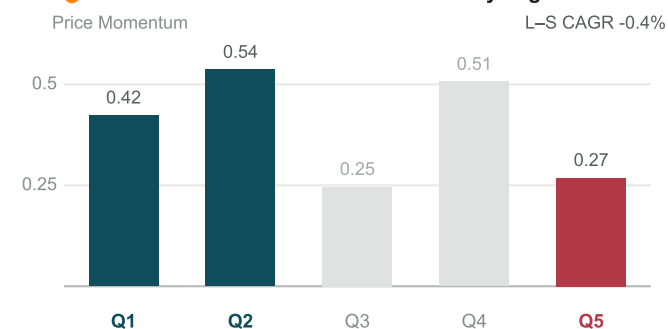
206 ● 26W Relative Price Strength



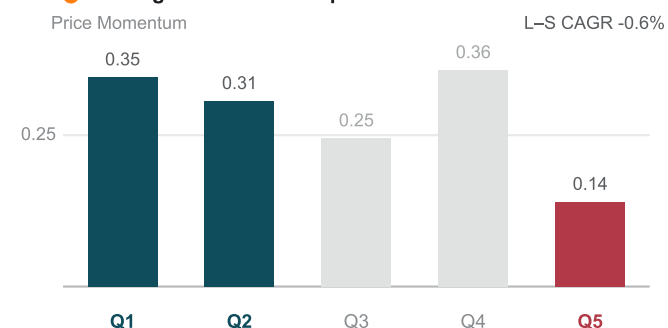
207 ● Change in Number of Holders



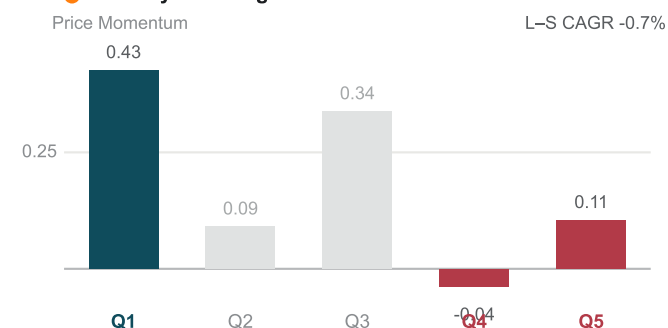
208 ● 52-Week Volume Price Trend with 20-Day Lag

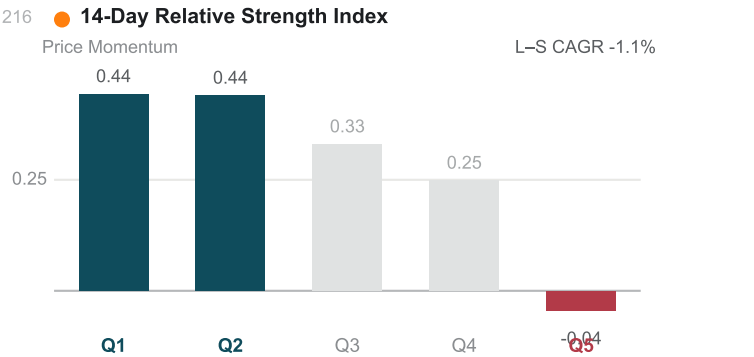
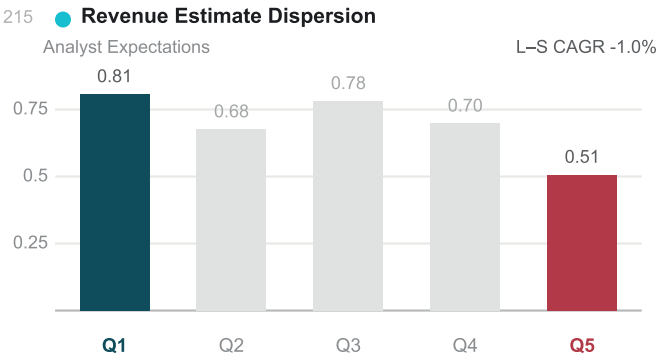
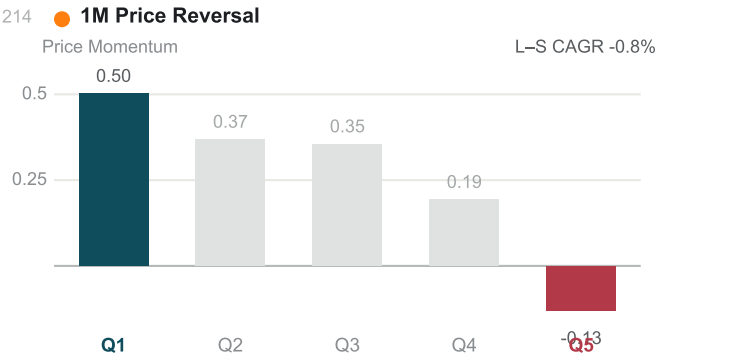
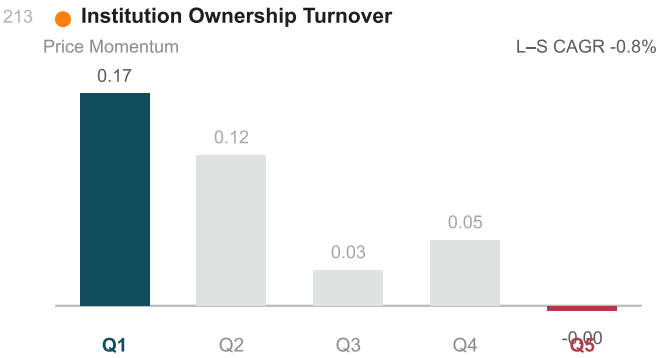
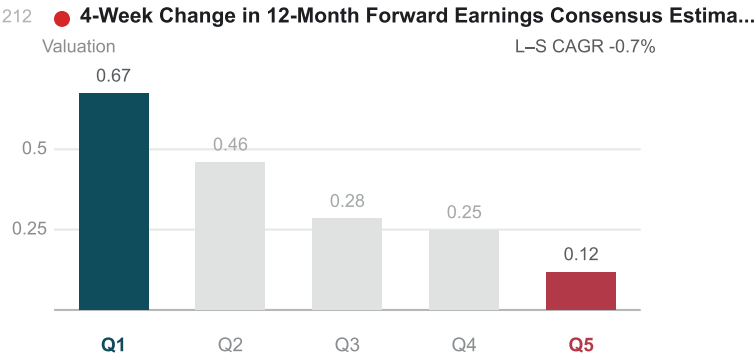
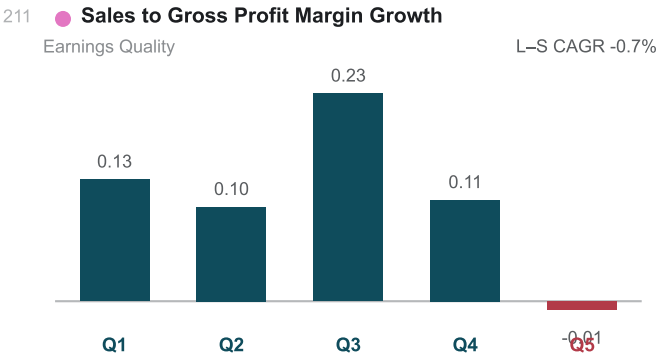


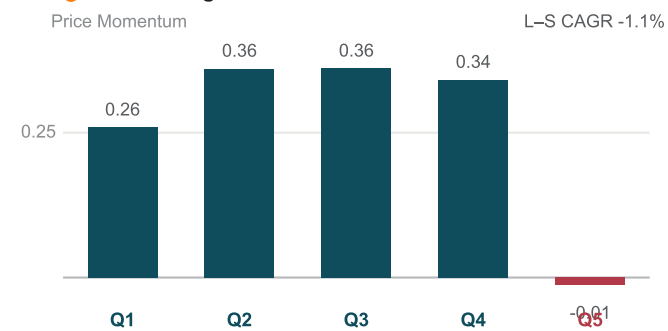
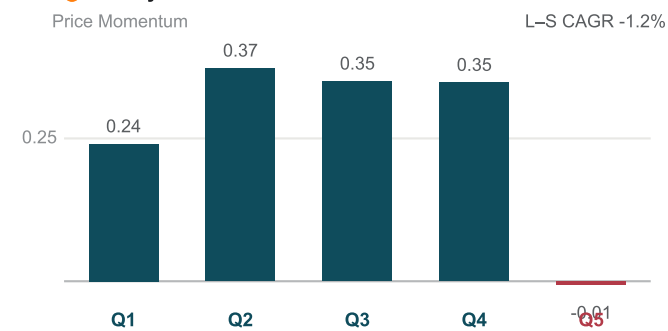
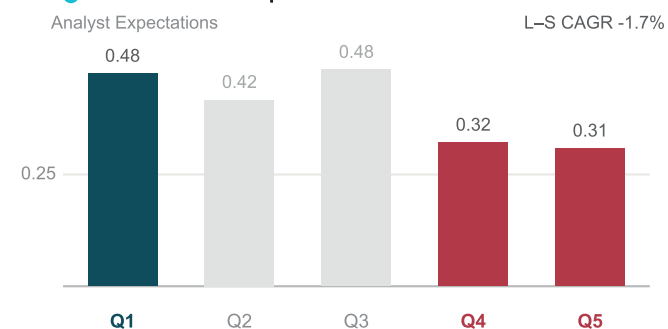
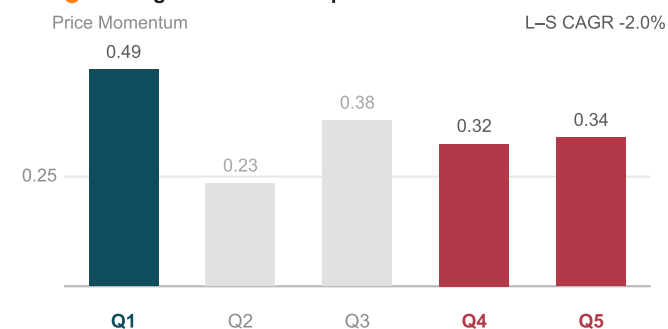
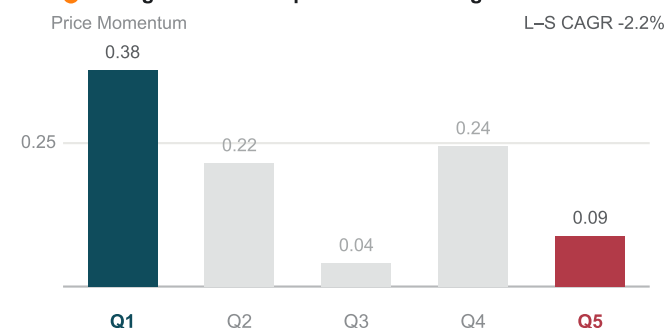
209 ● 6M Chg in 36M CAPM Alpha



210 ● Stability of Change in Number of Holders





217 ● **1M Price High - 1M Price Low**218 ● **20-Day Lane's Stochastic Indicator**219 ● **EBIT Estimate Dispersion**220 ● **6M Chg in 18M CAPM Alpha**221 ● **Change of Ownership Level - All Managers**222 ● **5-Day Money Flow/Volume**